

**All of the
Practicals is
performed and
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Github link:

<https://github.com/MakwanaMohit/MADPracticals>

Practical-1

AIM: Install Android Studio, set up the Android development environment, and create a simple "Hello World" app.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:layout_gravity="center"/>
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

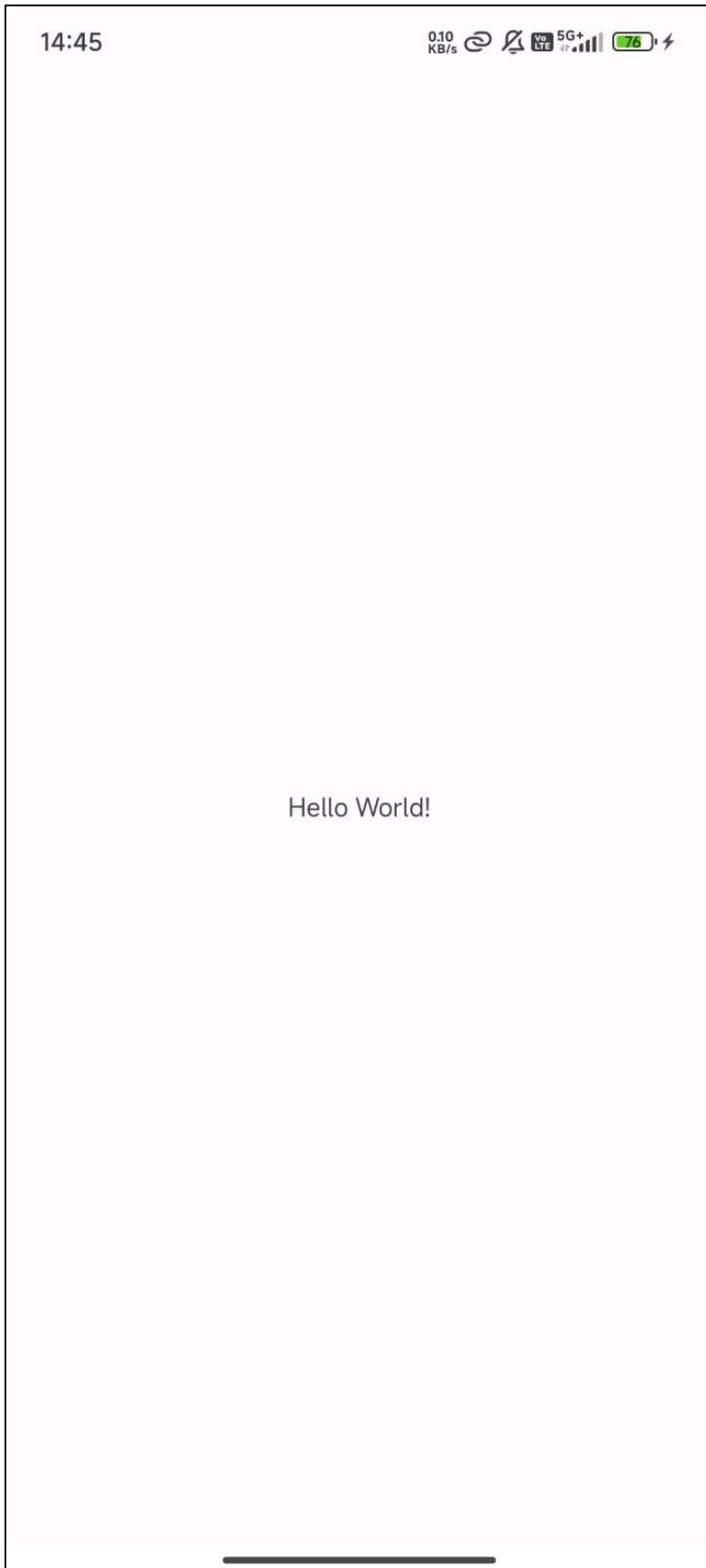
import android.os.Bundle;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical1 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical1);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        Toast.makeText(this, "welcome to Practical 1", Toast.LENGTH_SHORT).show();
    }
}
```

OUTPUT:



Practical-2

AIM: Develop a simple app that demonstrates the activity lifecycle.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:layout_gravity="center"/>
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.os.Handler;
import android.util.Log;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical2);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        Log.d("PRACTICAL2", "On Create called");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("PRACTICAL2", "On Start called");
    }
}
```

```

}

@Override
protected void onResume() {
    super.onResume();
    Log.d("PRACTICAL2", "On Resume called");
}

@Override
protected void onPause() {
    super.onPause();
    Log.d("PRACTICAL2", "On Pause called");
}

@Override
protected void onStop() {
    super.onStop();
    Log.d("PRACTICAL2", "On Stop called");
}

@Override
protected void onRestart() {
    super.onRestart();
    Log.d("PRACTICAL2", "On Restart called");
}

@Override
protected void onDestroy() {
    super.onDestroy();
    Log.d("PRACTICAL2", "On destroy called");
}
}

```

OUTPUT

2024-09-12 14:41:47.344 23833-23833 PRACTICAL2	com.mk.madpractical	D On Create called
2024-09-12 14:41:47.350 23833-23833 PRACTICAL2	com.mk.madpractical	D On Start called
2024-09-12 14:41:47.353 23833-23833 PRACTICAL2	com.mk.madpractical	D On Resume called
2024-09-12 14:41:47.406 23833-23833 VRI[Practical2]	com.mk.madpractical	D vri.reportNextDraw android.view.ViewRootImpl.performTraversals:4293 android.view.ViewRoo
2024-09-12 14:41:47.406 23833-23833 VRI[Practical2]	com.mk.madpractical	D vri.Setup new sync=wmsSync-VRI[Practical2]#2
2024-09-12 14:41:47.414 23833-23833 VRI[Practical2]	com.mk.madpractical	D vri.reportDrawFinished
2024-09-12 14:41:54.745 23833-23833 MIUIInput	com.mk.madpractical	D [KeyEvent] ViewRootImpl windowName 'sendBackKeyEvent-com.mk.madpractical/com.mk.madpract
2024-09-12 14:41:54.749 23833-23833 MIUIInput	com.mk.madpractical	D [KeyEvent] ViewRootImpl windowName 'sendBackKeyEvent-com.mk.madpractical/com.mk.madpract
2024-09-12 14:41:54.774 23833-23833 PRACTICAL2	com.mk.madpractical	D On Pause called
2024-09-12 14:41:55.417 23833-23833 PRACTICAL2	com.mk.madpractical	D On Stop called
2024-09-12 14:41:55.421 23833-23833 PRACTICAL2	com.mk.madpractical	D On destroy called

Logcat Output

Practical-3

AIM: Develop a simple calculator app that takes user input and performs basic arithmetic operations like addition, subtraction, multiplication, and division.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp"
    android:gravity="center">

    <EditText
        android:id="@+id/result"
        android:inputType="text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Result"
    />

    <EditText
        android:id="@+id/edt_1"
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
    />

    <EditText
        android:id="@+id/edt_2"
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
    />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center">

        <Button
            android:id="@+id/btn_plus"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="+"
            android:textSize="30sp"/>
```

```

<Button
    android:id="@+id/btn_minus"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="-"
    android:textSize="30sp"/>

<Button
    android:id="@+id/btn_multi"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="x"
    android:textSize="30sp"/>
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center">
    <Button
        android:id="@+id/btn_divi"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="/"
        android:textSize="30sp"/>
    <Button
        android:id="@+id/btn_modu"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="%"
        android:textSize="30sp"/>

</LinearLayout>
</LinearLayout>

```

JAVA:

```

package com.mk.madpractical;
import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;

```

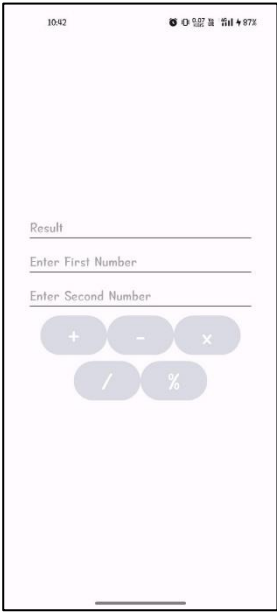


```

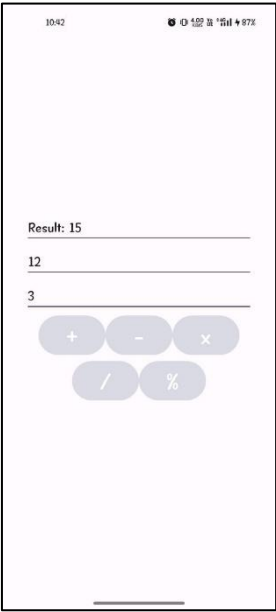
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class Practical3 extends AppCompatActivity {
    EditText n1, n2, Result;
    Button plus, minus, multi, division, modulus;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical3);
        n1 = (EditText) findViewById(R.id.edt_1);
        n2 = (EditText) findViewById(R.id.edt_2);
        plus = (Button) findViewById(R.id.btn_plus);
        minus = (Button) findViewById(R.id.btn_minus);
        multi = (Button) findViewById(R.id.btn_multi);
        division = (Button) findViewById(R.id.btn_divi);
        modulus = (Button) findViewById(R.id.btn_modu);
        Result = (EditText) findViewById(R.id.result);
        View.OnClickListener listener = new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int num1 = 0, num2 = 0, ans = 0;
                try {
                    num1 = Integer.parseInt(n1.getText().toString());
                    num2 = Integer.parseInt(n2.getText().toString());
                } catch (NumberFormatException e) {
                    Toast.makeText(Practical3.this, "Please enter a valid number for both", Toast.LENGTH_SHORT).show();
                }
                if (v.equals(plus)) {
                    ans = num1 + num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(minus)) {
                    ans = num1 - num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(multi)) {
                    ans = num1 * num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(division)) {
                    ans = num1 / num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(modulus)) {
                    ans = num1 % num2;
                    Result.setText("Result: " + ans);
                }
            }
        };
        plus.setOnClickListener(listener);
        minus.setOnClickListener(listener);
        multi.setOnClickListener(listener);
        division.setOnClickListener(listener);
        modulus.setOnClickListener(listener);
    }
}

```

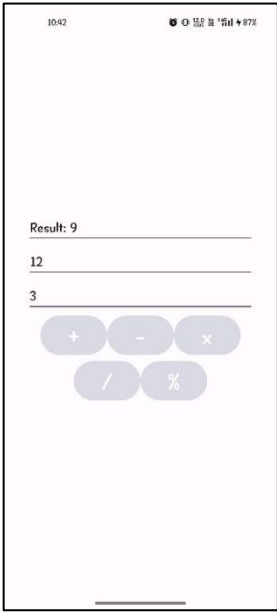
OUTPUT:



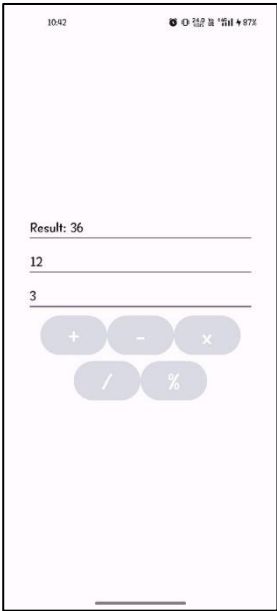
Activity layout



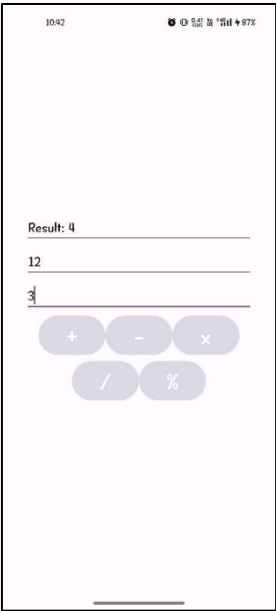
Plus operation



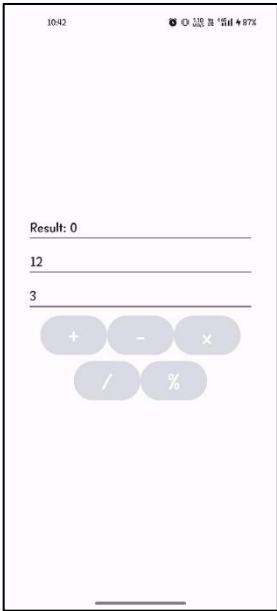
minus operation



Multiplication operation



Division operation



Modulus operation

Practical-4

AIM: Develop an Android application that uses LinearLayout to arrange UI components vertically or horizontally

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

    android:id="@+id/Practical_4_LinearLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 1"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 2"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 3"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 1"
            android:layout_margin="10dp"/>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 2"
            android:layout_margin="10dp"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 3"
            android:layout_margin="10dp"/>

    </LinearLayout>

</LinearLayout>

</LinearLayout>
```

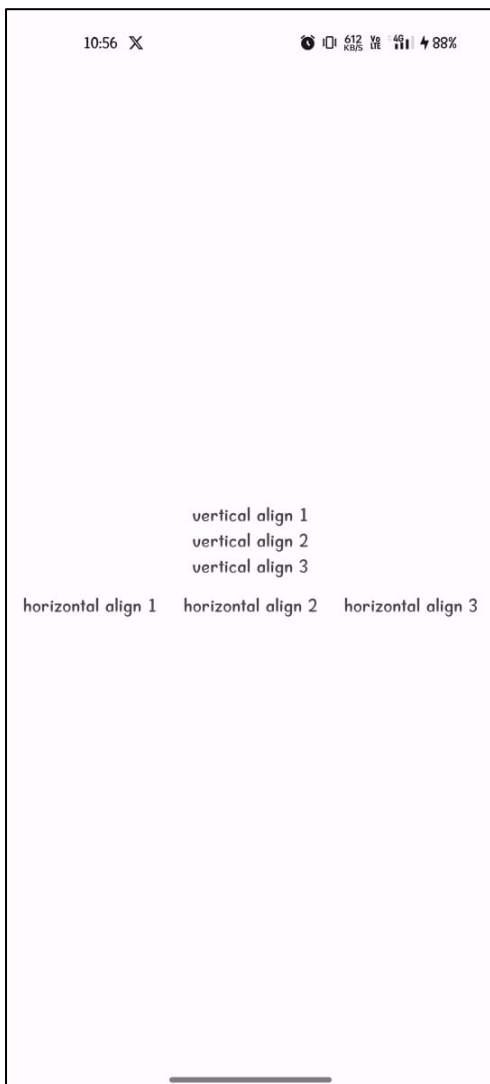
```

JAVA: package com.mk.madpractical;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical4 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical4);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical_4_LinearLayout), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}

```

OUTPUT



Practical-5

AIM: Develop an Android application that uses RelativeLayout to arrange UI components relative to each other.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_width="match_parent"
    android:id="@+id/Practical_5_RelativeLayout"
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <TextView
        android:id="@+id/centerText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Center text view" />
    <TextView
        android:id="@+id/topText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_above="@+id/centerText"
        android:layout_marginBottom="25dp"
        android:layout_centerHorizontal="true"
        android:text="Top of center text view" />
    <TextView
        android:id="@+id/Bottom_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/centerText"
        android:layout_marginTop="25dp"
        android:layout_centerHorizontal="true"
        android:text="Bottom of center text view" />
    <TextView
        android:id="@+id/Left_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="30dp"
        android:layout_centerVertical="true"
        android:layout_toStartOf="@+id/centerText"
        android:text="Left of text view" />
    <TextView
        android:id="@+id/Right_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="30dp"
        android:layout_centerVertical="true"
        android:layout_toEndOf="@+id/centerText"
        android:text="Right of text view" />

</RelativeLayout>
```

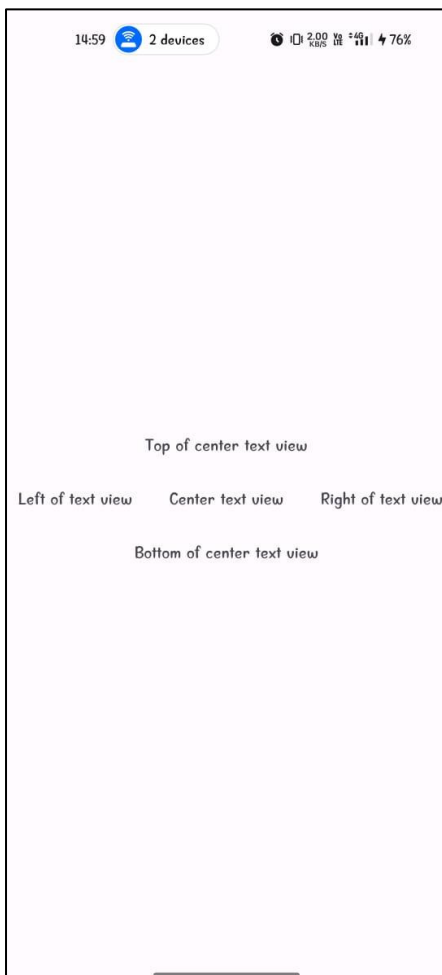
JAVA:

```
package com.mk.madpractical;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical5 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical5);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical_5_RelativeLayout), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

OUTPUT



Custom view for practical 6

XML: `<?xml version="1.0" encoding="utf-8"?>`

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/My_Custom_View_Layout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">
    <ImageView
        android:id="@+id/customImage"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_gravity="center_vertical"
        android:src="@drawable/ic_launcher_foreground"
        android:scaleType="centerCrop" />
    <TextView
        android:id="@+id/customText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_gravity="center_vertical"
        android:text="Custom test here"
        android:layout_marginStart="10dp" />
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Rect;
import android.util.AttributeSet;
import android.view.LayoutInflater;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.TextView;

import java.util.Random;

public class MyCustomView extends LinearLayout {

    private TextView customText;
    private ImageView customImage;

    public MyCustomView(Context context) {
        super(context);
        init(context);
    }
}
```

```

public MyCustomView(Context context, AttributeSet attrs) {
    super(context, attrs);
    init(context);
}

public MyCustomView(Context context, AttributeSet attrs, int defStyleAttr) {
    super(context, attrs, defStyleAttr);
    init(context);
}

private void init(Context context) {
    LayoutInflater inflater = LayoutInflater.from(context);
    inflater.inflate(R.layout.custom_layout, this, true);
    customText = findViewById(R.id.customText);
    customImage = findViewById(R.id.customImage);
}

public void setCustomText(String text) {
    customText.setText(text);
}

public void setCustomImage(int resourcelId) {
    customImage.setImageResource(resourcelId);
}

public void setCustomImage (Bitmap bitmap){
    customImage.setImageBitmap(bitmap);
}

public Bitmap createBitmapWithLetter(String letter, int color) {
    if (letter.length() > 3){
        letter = letter.substring(0,3);
    }
    Bitmap bitmap = Bitmap.createBitmap(100, 100, Bitmap.Config.ARGB_8888);
    Canvas canvas = new Canvas(bitmap);
    canvas.drawColor(color);
    Paint paint = new Paint();
    paint.setColor(Color.WHITE);
    paint.setTextSize(50);
    paint.setTextAlign(Paint.Align.CENTER);

    Rect bounds = new Rect();
    paint.getTextBounds(letter, 0, letter.length(), bounds);
    int x = (bitmap.getWidth() / 2);
    int y = (int) ((bitmap.getHeight() / 2) - ((paint.descent() + paint.ascent()) / 2));

    canvas.drawText(letter, x, y, paint);

    return bitmap;
}

public int getRandomColor() {
    Random random = new Random();
    return Color.rgb(random.nextInt(256), random.nextInt(256), random.nextInt(256));
}
}

```


Practical-6

AIM: Develop an Android application that uses ScrollView to display a long list of items.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical6">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:id="@+id/Pr6_ScrollView_LinearLayout"
            android:orientation="vertical" />
    </ScrollView>
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.view.Gravity;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.LinearLayout;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical6 extends AppCompatActivity {
    LinearLayout layout;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

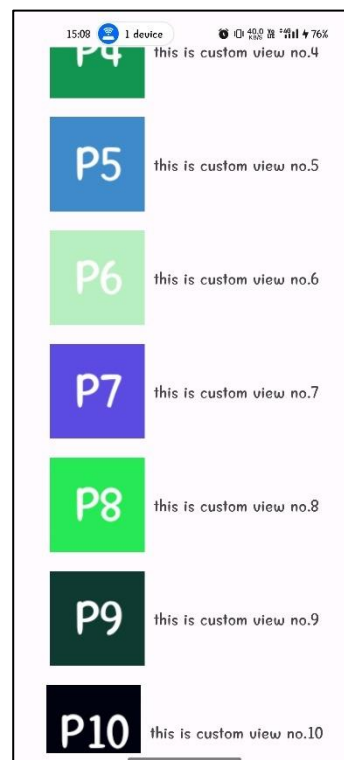
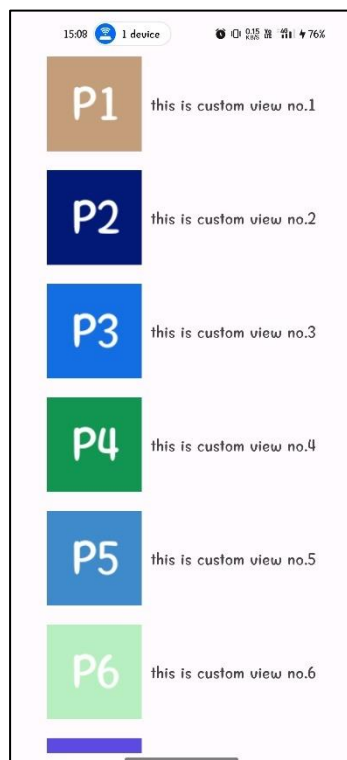
```

EdgeToEdge.enable(this);
setContentView(R.layout.activity_practical6);
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});
layout = findViewById(R.id.Pr6_ScrollView_LinearLayout);
for (int i = 1; i < 31; i++) {
    MyCustomView view = new MyCustomView(this);
    view.setCustomImage(view.createBitmapWithLetter("P"+i,view.getRandomColor()));
    view.setCustomText("this is custom view no."+i);
    LinearLayout lay = (LinearLayout) view.findViewById(R.id.My_Custom_View_Layout);
    lay.setGravity(Gravity.CENTER);
    ImageView imageView = view.findViewById(R.id.customImage);
    int newWidthInDp = 100;
    int newHeightInDp = 100;

    float scale = getResources().getDisplayMetrics().density;
    int newWidthInPx = (int) (newWidthInDp * scale + 0.5f);
    int newHeightInPx = (int) (newHeightInDp * scale + 0.5f);

    ViewGroup.LayoutParams params = imageView.getLayoutParams();
    params.width = newWidthInPx;
    params.height = newHeightInPx;
    imageView.setLayoutParams(params);
    layout.addView(view);
}
}
}

```



Custom view for practical 7

XML: `<?xml version="1.0" encoding="utf-8"?>`

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/My_Custom_View_Layout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">
    <ImageView
        android:id="@+id/customImage"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_gravity="center_vertical"
        android:src="@drawable/ic_launcher_foreground"
        android:scaleType="centerCrop" />
    <TextView
        android:id="@+id/customText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_gravity="center_vertical"
        android:text="Custom test here"
        android:layout_marginStart="10dp" />
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Rect;
import android.util.AttributeSet;
import android.view.LayoutInflater;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.TextView;

import java.util.Random;

public class MyCustomView extends LinearLayout {

    private TextView customText;
    private ImageView customImage;

    public MyCustomView(Context context) {
        super(context);
        init(context);
    }
}
```

```

public MyCustomView(Context context, AttributeSet attrs) {
    super(context, attrs);
    init(context);
}

public MyCustomView(Context context, AttributeSet attrs, int defStyleAttr) {
    super(context, attrs, defStyleAttr);
    init(context);
}

private void init(Context context) {
    LayoutInflater inflater = LayoutInflater.from(context);
    inflater.inflate(R.layout.custom_layout, this, true);
    customText = findViewById(R.id.customText);
    customImage = findViewById(R.id.customImage);
}

public void setCustomText(String text) {
    customText.setText(text);
}

public void setCustomImage(int resourceId) {
    customImage.setImageResource(resourceId);
}

public void setCustomImage (Bitmap bitmap){
    customImage.setImageBitmap(bitmap);
}

public Bitmap createBitmapWithLetter(String letter, int color) {
    if (letter.length() > 3){
        letter = letter.substring(0,3);
    }
    Bitmap bitmap = Bitmap.createBitmap(100, 100, Bitmap.Config.ARGB_8888);
    Canvas canvas = new Canvas(bitmap);
    canvas.drawColor(color);
    Paint paint = new Paint();
    paint.setColor(Color.WHITE);
    paint.setTextSize(50);
    paint.setTextAlign(Paint.Align.CENTER);

    Rect bounds = new Rect();
    paint.getTextBounds(letter, 0, letter.length(), bounds);
    int x = (bitmap.getWidth() / 2);
    int y = (int) ((bitmap.getHeight() / 2) - ((paint.descent() + paint.ascent()) / 2));

    canvas.drawText(letter, x, y, paint);

    return bitmap;
}

public int getRandomColor() {
    Random random = new Random();
    return Color.rgb(random.nextInt(256), random.nextInt(256), random.nextInt(256));
}
}

```

Practical-7

AIM: Develop an Android application that uses ListView and Custom Adapter to display a list of images with text.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical7">
    <ListView
        android:id="@+id/PR7_ListView"
        android:layout_width="300dp"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;
import android.content.Context; import android.graphics.*; import android.os.Bundle;
import android.view.LayoutInflater; import android.view.View; import android.view.ViewGroup;
import android.widget.BaseAdapter; import android.widget.ImageView; import android.widget.ListView;
import android.widget.TextView; import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import java.util.Random;

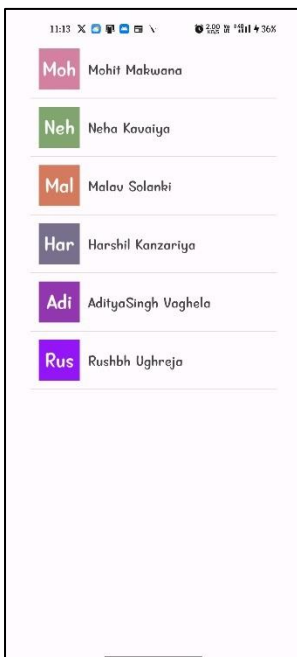
public class Practical7 extends AppCompatActivity {
    ListView listView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical7);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        listView = findViewById(R.id.PR7_ListView);
        String[] arr = {"Mohit Makwana", "Neha Kavaia", "Malav Solanki", "Harshil Kanzariya", "AdityaSingh Vaghela",
        "Rushbh Ughreja"};
        listView.setAdapter(new Pr7CustomAdapter(this, arr));
    }
}
```

```

class Pr7CustomAdapter extends BaseAdapter {
    private Context context;
    private String[] items;
    public Pr7CustomAdapter(Context context, String[] items) {
        this.context = context;
        this.items = items;
    }
    @Override
    public int getCount() {
        return items.length;
    }
    @Override
    public Object getItem(int position) {
        return items[position];
    }
    @Override
    public long getItemId(int position) {
        return position;
    }
    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        MyCustomView view;
        if (convertView == null) {
            view = new MyCustomView(context);
        } else {
            view = (MyCustomView) convertView;
        }
        String item = (String) getItem(position);
        view.setCustomText(item);
        view.setCustomImage(view.createBitmapWithLetter(item, view.getRandomColor()));
        return (View) view;
    }
}

```

OUTPUT



Practical-8

AIM: Develop an Android application that uses the Navigation Drawer to display a side menu.

XML:

Practical_layout file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:openDrawer="start">

    <include
        android:id="@+id/app_bar_practical8"
        layout="@layout/app_bar_practical8"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        android:fitsSystemWindows="true"
        app:headerLayout="@layout/nav_header_practical8"
        app:menu="@menu/activity_main_drawer" />
</androidx.drawerlayout.widget.DrawerLayout>
```

App_bar_layout:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:openDrawer="start">

    <include
        android:id="@+id/app_bar_practical8"
        layout="@layout/app_bar_practical8"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
```

```
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        android:fitsSystemWindows="true"
        app:headerLayout="@layout/nav_header_practical8"
        app:menu="@menu/activity_main_drawer" />
</androidx.drawerlayout.widget.DrawerLayout>
```

content_layout:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

    app:layout_behavior="@string/appbar_scrolling_view_behavi
or"
    tools:showIn="@layout/app_bar_practical8">

    <fragment

        android:id="@+id/nav_host_fragment_content_practical8"

        android:name="androidx.navigation.fragment.NavHostFragme
nt"

        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:defaultNavHost="true"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:navGraph="@navigation/mobile_navigation" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Home_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.home.HomeFragment">

    <TextView
        android:id="@+id/text_home"
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Gallery_fragment:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.gallery.GalleryFragment">

    <TextView
        android:id="@+id/text_gallery"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

SlidShow_fragment:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.slideshow.SlideshowFragment">

    <TextView
        android:id="@+id/text_slideshow"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"

```

```

        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Activity_main_drawer:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    tools:showIn="navigation_view">

    <group android:checkableBehavior="single">
        <item
            android:id="@+id/nav_home"
            android:icon="@drawable/ic_menu_camera"
            android:title="@string/menu_home" />
        <item
            android:id="@+id/nav_gallery"
            android:icon="@drawable/ic_menu_gallery"
            android:title="@string/menu_gallery" />
        <item
            android:id="@+id/nav_slideshow"
            android:icon="@drawable/ic_menu_slideshow"
            android:title="@string/menu_slideshow" />
    </group>
</menu>

```

Movile_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<navigation
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/mobile_navigation"
    app:startDestination="@+id/nav_home">
    <fragment
        android:id="@+id/nav_home"
        android:name="com.mk.madpractical.ui.home.HomeFragment"
        android:label="@string/menu_home"
        tools:layout="@layout/fragment_home" />
    <fragment
        android:id="@+id/nav_gallery"
        android:name="com.mk.madpractical.ui.gallery.GalleryFragment"
        android:label="@string/menu_gallery"
        tools:layout="@layout/fragment_gallery" />
</navigation>

```



```

        android:id="@+id/nav_slideshow"

        android:name="com.mk.madpractical.ui.slideshow.SlideshowF
        ragment"
        android:label="@string/menu_slideshow"
        tools:layout="@layout/fragment_slideshow" />
    </navigation>

```

Nav_header_layout:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="@dimen/nav_header_height"
    android:background="@drawable/side_nav_bar"
    android:gravity="bottom"
    android:orientation="vertical"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:theme="@style/ThemeOverlay.AppCompat.Dark">
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:contentDescription="@string/nav_header_desc"

        android:paddingTop="@dimen/nav_header_vertical_spacing"
        app:srcCompat="@mipmap/ic_launcher_round" />
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:paddingTop="@dimen/nav_header_vertical_spacing"
        android:text="@string/nav_header_title"

        android:textAppearance="@style/TextAppearance.AppCompat
        .Body1" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/nav_header_subtitle" />
</LinearLayout>

```

Practical_8_menu_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/action_settings"

```

```

        android:orderInCategory="100"
        android:title="@string/action_settings"
        app:showAsAction="never" />
</menu>

```

JAVA:

Home_fragment & Home_view_model :

```

package com.mk.madpractical.ui.home;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import com.mk.madpractical.databinding.FragmentHomeBinding;
public class HomeFragment extends Fragment {
    private FragmentHomeBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
        savedInstanceState) {
        HomeViewModel homeViewModel =
            new
            ViewModelProvider(this).get(HomeViewModel.class);
        binding = FragmentHomeBinding.inflate(inflater,
            container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textHome;

        homeViewModel.getText().observe(getViewLifecycleOwner(),
            textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class HomeViewModel extends ViewModel {

    private final MutableLiveData<String> mText;

    public HomeViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is home fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Gallery_fragment & Gallery_view_model :

```

package com.mk.madpractical.ui.gallery;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentGalleryBinding;
public class GalleryFragment extends Fragment {
    private FragmentGalleryBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        GalleryViewModel galleryViewModel =
            new
ViewModelProvider(this).get(GalleryViewModel.class);
        binding = FragmentGalleryBinding.inflate(inflater,
container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textGallery;

galleryViewModel.getText().observe(getViewLifecycleOwner(),

textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class GalleryViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public GalleryViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is gallery fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Slideshow_fragment & / Slideshow_view_model :

```

package com.mk.madpractical.ui.slideshow;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

```

```

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentSlideshowBinding
;
public class SlideshowFragment extends Fragment {
    private FragmentSlideshowBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        SlideshowViewModel slideshowViewModel =
            new
ViewModelProvider(this).get(SlideshowViewModel.class);
        binding = FragmentSlideshowBinding.inflate(inflater,
container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textSlideshow;

slideshowViewModel.getText().observe(getViewLifecycleOwner(),

textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class SlideshowViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public SlideshowViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is slideshow fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Practical_8.java :

```

package com.mk.madpractical;
import android.os.Bundle;
import android.view.View;
import android.view.Menu;
import com.google.android.material.snackbar.Snackbar;
import
com.google.android.material.navigation.NavigationView;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;

```

```

import androidx.drawerlayout.widget.DrawerLayout;
import androidx.appcompat.app.AppCompatActivity;
import
com.mk.madpractical.databinding.ActivityPractical8Binding;
public class Practical8 extends AppCompatActivity {
    private AppBarConfiguration mAppBarConfiguration;
    private ActivityPractical8Binding binding;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        binding =
        ActivityPractical8Binding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        setSupportActionBar(binding.appBarPractical8.toolbar);
        binding.appBarPractical8.fab.setOnClickListener(new
        View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Snackbar.make(view, "Message function will
                comming soon", Snackbar.LENGTH_LONG)
                    .setAction("Action", null)
                    .setAnchorView(R.id.fab).show();
            }
        });
        DrawerLayout drawer = binding.drawerLayout;
        NavigationView navigationView =
binding.Practical8NavigationView;
        // Passing each menu ID as a set of Ids because each
        // menu should be considered as top level destinations.

```

```

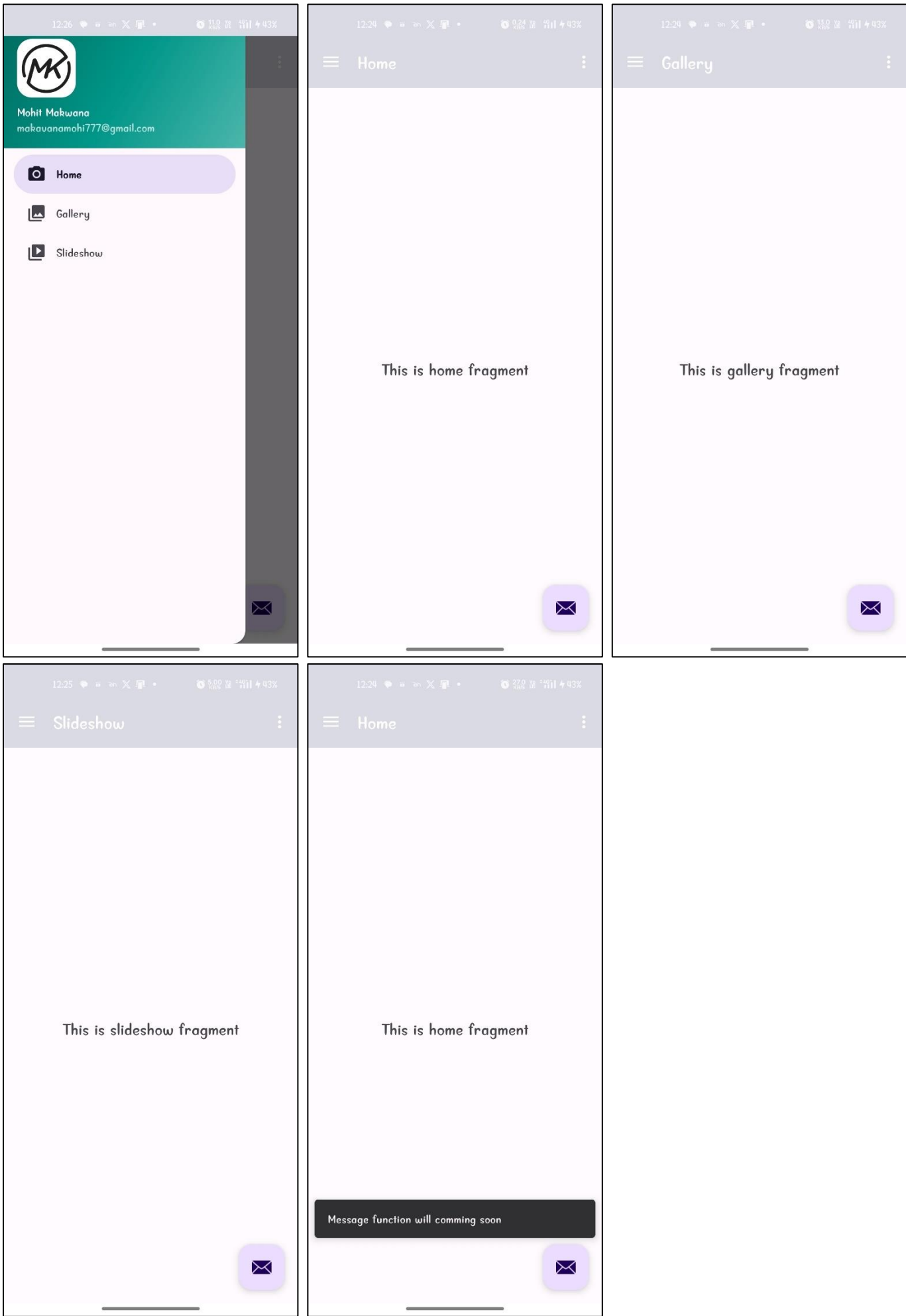
        mAppBarConfiguration = new
        AppBarConfiguration.Builder(
            R.id.nav_home, R.id.nav_gallery, R.id.nav_slideshow)
            .setOpenableLayout(drawer)
            .build();
        NavController navController =
        Navigation.findNavController(this,
        R.id.nav_host_fragment_content_practical8);
        NavigationUI.setupActionBarWithNavController(this,
        navController, mAppBarConfiguration);
        NavigationUI.setupWithNavController(navigationView,
        navController);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is
        present.
        getMenuInflater().inflate(R.menu.practical8, menu);
        return true;
    }

    @Override
    public boolean onSupportNavigateUp() {
        NavController navController =
        Navigation.findNavController(this,
        R.id.nav_host_fragment_content_practical8);
        return NavigationUI.navigateUp(navController,
mAppBarConfiguration)
            || super.onSupportNavigateUp();
    }
}

```

OUTPUT:



Practical-9

AIM: Develop an Android application that uses the bottom navigation bar to switch between different tabs.

XML:

Practical_layout file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingTop="?attr/actionBarSize">
    <com.google.android.material.bottomnavigation.
```

BottomNavigationView

```
    android:id="@+id/nav_view"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="0dp"
    android:layout_marginEnd="0dp"
    android:background="?android:attr/windowBackground"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:menu="@menu/bottom_nav_menu" />
</fragment
    android:id="@+id/nav_host_fragment_activity_practical9"
```

```
android:name="androidx.navigation.fragment.NavHostFragment"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:defaultNavHost="true"
    app:layout_constraintBottom_toTopOf="@id/nav_view"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:navGraph="@navigation/mobile_navigation3" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Home_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.home.HomeFragment">
    <TextView
        android:id="@+id/text_home"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Dashboard_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.dashboard.DashboardFragment">
    <TextView
        android:id="@+id/text_dashboard"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Notifications_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.notifications.NotificationsFragment">
    <TextView
        android:id="@+id/text_notifications"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
```

```

        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Bottom_nav_menu:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
xmlns:android="http://schemas.android.com/apk/res/android"
>
    <item
        android:id="@+id/navigation_home"
        android:icon="@drawable/ic_home_black_24dp"
        android:title="@string/title_home" />
    <item
        android:id="@+id/navigation_dashboard"
        android:icon="@drawable/ic_dashboard_black_24dp"
        android:title="@string/title_dashboard" />
    <item
        android:id="@+id/navigation_notifications"
        android:icon="@drawable/ic_notifications_black_24dp"
        android:title="@string/title_notifications" />
</menu>

```

Mobile_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<navigation
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/mobile_navigation3"
app:startDestination="@+id/navigation_home">
    <fragment
        android:id="@+id/navigation_home"
        android:name="com.mk.madpractical.ui.home.HomeFragment"
        android:label="@string/title_home"
        tools:layout="@layout/fragment_home" />
    <fragment
        android:id="@+id/navigation_dashboard"
        android:name="com.mk.madpractical.ui.dashboard.DashboardFragment"
        android:label="@string/title_dashboard"
        tools:layout="@layout/fragment_dashboard" />
    <fragment
        android:id="@+id/navigation_notifications"
        android:name="com.mk.madpractical.ui.notifications.NotificationsFragment"
        android:label="@string/title_notifications"
        tools:layout="@layout/fragment_notifications" />
</navigation>

```

JAVA:

Practical_9.java :

```

package com.mk.madpractical;
import android.os.Bundle;
import
com.google.android.material.bottomnavigation.BottomNavigationView;
import androidx.appcompat.app.AppCompatActivity;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;
import
com.mk.madpractical.databinding.ActivityPractical9Binding;

public class Practical9 extends AppCompatActivity {
    private ActivityPractical9Binding binding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        binding =
ActivityPractical9Binding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());
        BottomNavigationView navView =
findViewById(R.id.nav_view);
        // Passing each menu ID as a set of IDs because each
        // menu should be considered as top level destinations.
        AppBarConfiguration appBarConfiguration = new
AppBarConfiguration.Builder(
            R.id.navigation_home, R.id.navigation_dashboard,
            R.id.navigation_notifications)
            .build();
        NavController navController =
Navigation.findNavController(this,
            R.id.nav_host_fragment_activity_practical9);
        NavigationUI.setupActionBarWithNavController(this,
            navController, appBarConfiguration);
        NavigationUI.setupWithNavController(binding.navView,
            navController);
    }
}

```

Home_fragment & Home_view_model :

```

package com.mk.madpractical.ui.home;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;

```

```

import
com.mk.madpractical.databinding.FragmentHomeBinding;
public class HomeFragment extends Fragment {
    private FragmentHomeBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        HomeViewModel homeViewModel =
            new
ViewModelProvider(this).get(HomeViewModel.class);
        binding = FragmentHomeBinding.inflate(inflater,
container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textHome;
        homeViewModel.getText().observe(getViewLifecycleOwner(), te
xtView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
}class HomeViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public HomeViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is home fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}
}

```

Dashboard_fragment & Dashboard_view_model :

```

package com.mk.madpractical.ui.dashboard;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.ViewModelProvider;
import com.mk.madpractical.databinding.

```

```

FragmentDashboardBinding;

```

```

public class DashboardFragment extends Fragment {
    private FragmentDashboardBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        DashboardViewModel dashboardViewModel =
            new
ViewModelProvider(this).get(DashboardViewModel.class);

```

```

        binding = FragmentDashboardBinding.inflate(inflater,
container,
false);
        View root = binding.getRoot();
        final TextView textView = binding.textDashboard;

dashboardViewModel.getText().observe(getViewLifecycleOwn
er(), textView::setText);
        return root;
    } @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
}class DashboardViewModel extends ViewModel {

    private final MutableLiveData<String> mText;

    public DashboardViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is dashboard fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}
}

```

Notifications_fragment & Notifications_view_model :

```

package com.mk.madpractical.ui.notifications;`
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentNotificationsBindi
ng;
public class NotificationsFragment extends Fragment {
    private FragmentNotificationsBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        NotificationsViewModel notificationsViewModel =
            new
ViewModelProvider(this).get(NotificationsViewModel.class);
        binding = FragmentNotificationsBinding.inflate(inflater,
container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textNotifications;

notificationsViewModel.getText().observe(getViewLifecycleOw
ner(), textView::setText);
        return root;
    }
}

```

```

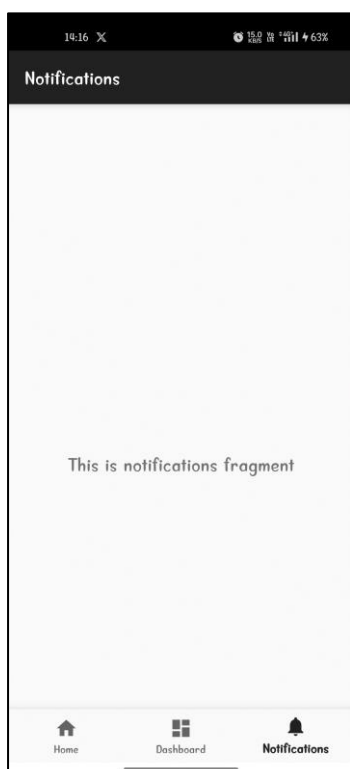
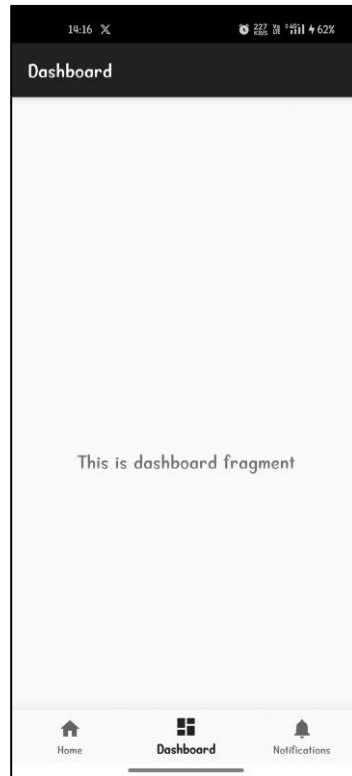
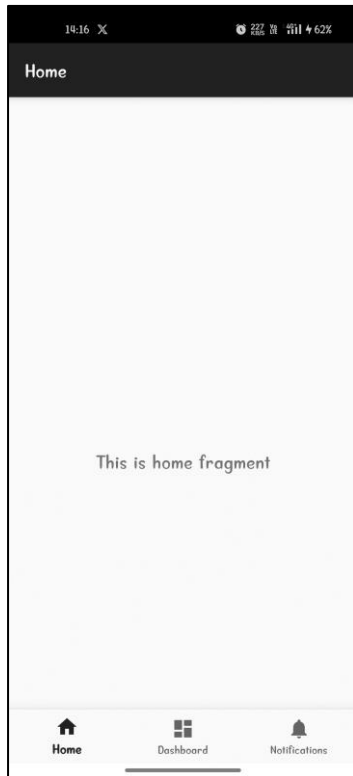
@Override
public void onDestroyView() {
    super.onDestroyView();
    binding = null;
}
}class NotificationsViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public NotificationsViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is notifications fragment");
    }
}

    }

    public LiveData<String> getText() {
        return mText;
    }
}

```

OUTPUT:



Practical-10

AIM: Develop an Android application that uses an Intent to pass data between different activities

XML:

MainActivity:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

</androidx.constraintlayout.widget.ConstraintLayout>
```

Practical-10:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical_10_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical10">
<TextView
    android:id="@+id/Practical_10_textView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

MainActivity:

```
package com.mk.madpractical;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.os.Bundle;
```

```
import android.app.AlertDialog;
import android.content.DialogInterface;
import android.widget.EditText;
import android.widget.LinearLayout;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());

            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        showInputDialog();
    }
    private void showInputDialog() {
        // Create EditText programmatically
        final EditText editTextInput = new EditText(this);
        editTextInput.setHint("Enter your text");
        editTextInput.setLayoutParams(new
        LinearLayout.LayoutParams(
            LinearLayout.LayoutParams.MATCH_PARENT,
            LinearLayout.LayoutParams.MATCH_PARENT));
        new AlertDialog.Builder(this)
            .setTitle("Enter your text")
            .setView(editTextInput)
            .setPositiveButton("OK", new
            DialogInterface.OnClickListener() {
                @Override
                public void onClick(DialogInterface dialog, int
                which) {
                    String inputText =
                    editTextInput.getText().toString();
                    Intent pr10 = new
                    Intent(MainActivity.this, Practical10.class);
                    pr10.putExtra("MSG", inputText);
                    startActivity(pr10);
                }
            })
            .setNegativeButton("Cancel", null)
            .show();
    }
}
```

```

}
@Override
protected void onDestroy() {
    super.onDestroy();
}
}

```

Practical-10:

```

package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.android.material.snackbar.Snackbar;

public class Practical10 extends AppCompatActivity {

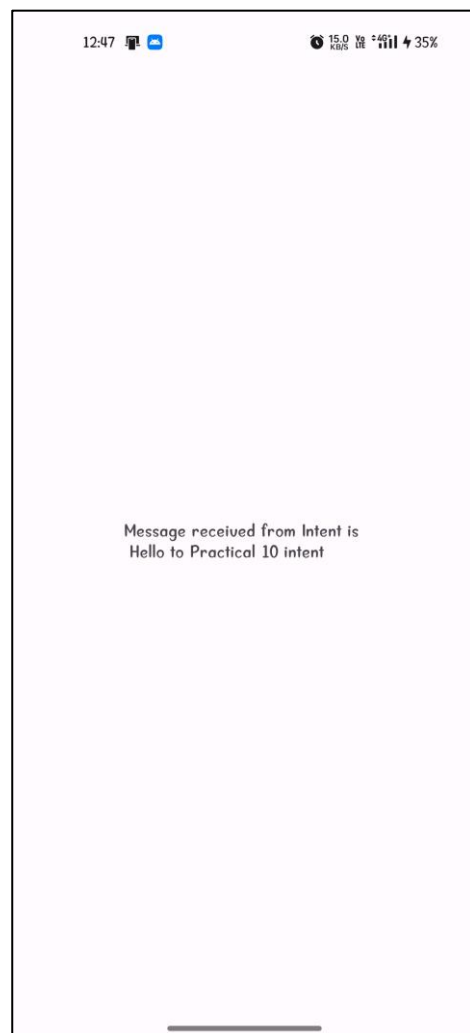
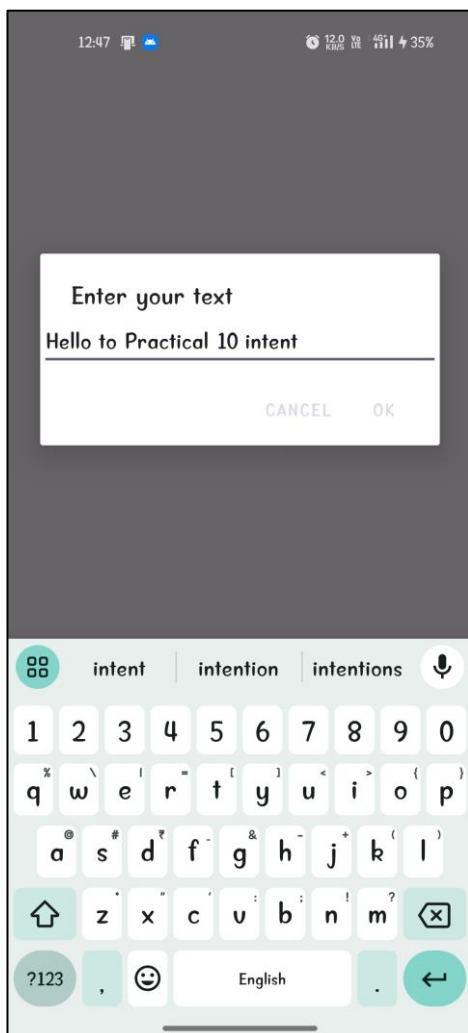
```

```

    private TextView textView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical10);

        ViewCompat.setOnApplyWindowInsetsListener(findView
        ById(R.id.Practical_10_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars()
            );
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        textView =
        findViewById(R.id.Practical_10_textView1);
        Intent intent = getIntent();
        textView.setText("Message received from Intent is
        \n "+intent.getStringExtra("MSG"));
    }
}

```



Practical-11

AIM: Develop an Android application that uses Services to perform background tasks.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical11_layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical11">
<Button
    android:id="@+id/Practical11_start_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start the service"

app:layout_constraintBottom_toTopOf="@id/Practical11_stop_btn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
<Button
    android:id="@+id/Practical11_stop_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Stop the service"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Practical11_start_btn" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Service.java

```
package com.mk.madpractical;

import android.app.Service;
import android.content.Intent;
import android.os.Binder;
import android.os.Handler;
import android.os.Looper;
```

```
import android.os.IBinder;
import android.widget.Toast;

public class Practical11Service extends Service {

    private Thread thread;
    private boolean isServiceRunning;

    public Practical11Service() {
    }

    @Override
    public IBinder onBind(Intent intent) {
        return new Binder() {
            Practical11Service get_service() {
                return Practical11Service.this;
            }
        };
    }

    @Override
    public void onCreate() {
        super.onCreate();
        isServiceRunning = true;

        // Create and start a new thread
        thread = new Thread(new Runnable() {
            @Override
            public void run() {
                while (isServiceRunning) {
                    new
                    Handler(Looper.getMainLooper()).post(new Runnable() {
                        @Override
                        public void run() {
                            Toast.makeText(Practical11Service.this,
                            "Service is running", Toast.LENGTH_SHORT).show();
                        }
                    });
                }
            }
        });
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int
```

```
startId) {
    if (!thread.isAlive()) {
        thread.start(); // Start the thread
    }
    return START_STICKY; // Keep the service running
}
```

```
@Override
public void onDestroy() {
    super.onDestroy();
    isServiceRunning = false; // Stop the loop
    if (thread != null && thread.isAlive()) {
        thread.interrupt(); // Safely stop the thread
    }
}
```

Practical11.java

```
package com.mk.madpractical;
```

```
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class Practical11 extends AppCompatActivity {
    Button start,stop;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
EdgeToEdge.enable(this);
setContentView(R.layout.activity_practical11);
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical11_layout), (v, insets) -> {
    Insets systemBars =
    insets.getInsets(WindowInsetsCompat.Type.systemBars())
});
```

```
v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
return insets;
});
```

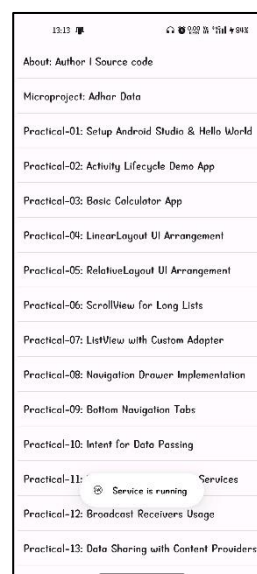
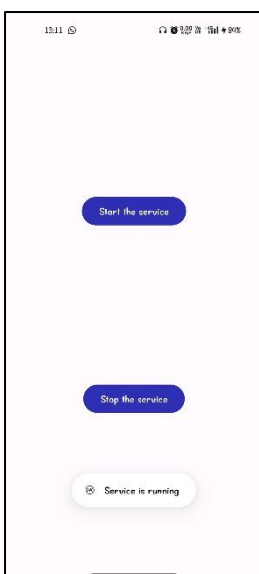
```
start = findViewById(R.id.Practical11_start_btn);
stop = findViewById(R.id.Practical11_stop_btn);
Intent service = new
Intent(this,Practical11Service.class);
```

```
View.OnClickListener listener = new
View.OnClickListener() {
```

```
@Override
public void onClick(View v) {
```

```
try {
    if (v.equals(start)){
        startService(service);
    }else {
        stopService(service);
    }
} catch (Exception e) {
    throw new RuntimeException(e);
}
}
```

```
};
start.setOnClickListener(listener);
stop.setOnClickListener(listener);
}
```



Practical-12

AIM: Develop an Android application that uses Broadcast Receivers to receive and handle system-level broadcasts.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical12_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical12_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="plug the power cable or take it off to
see the broadcast"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.IntentFilter;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
```

```
public class Practical12 extends AppCompatActivity {
    Practical12_broadcast broadcast;
    IntentFilter intentFilter;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical12);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical12_ConstraintLayout), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

        textView = findViewById(R.id.Practical12_textview);

        broadcast = new Practical12_broadcast();
        intentFilter = new IntentFilter();

        intentFilter.addAction("android.intent.action.ACTION_POWER_CONNECTED");

        intentFilter.addAction("android.intent.action.ACTION_POWER_DISCONNECTED");
        registerReceiver(broadcast, intentFilter,
Context.RECEIVER_NOT_EXPORTED);

    }

    @Override
    protected void onDestroy() {
        unregisterReceiver(broadcast);
        super.onDestroy();
    }

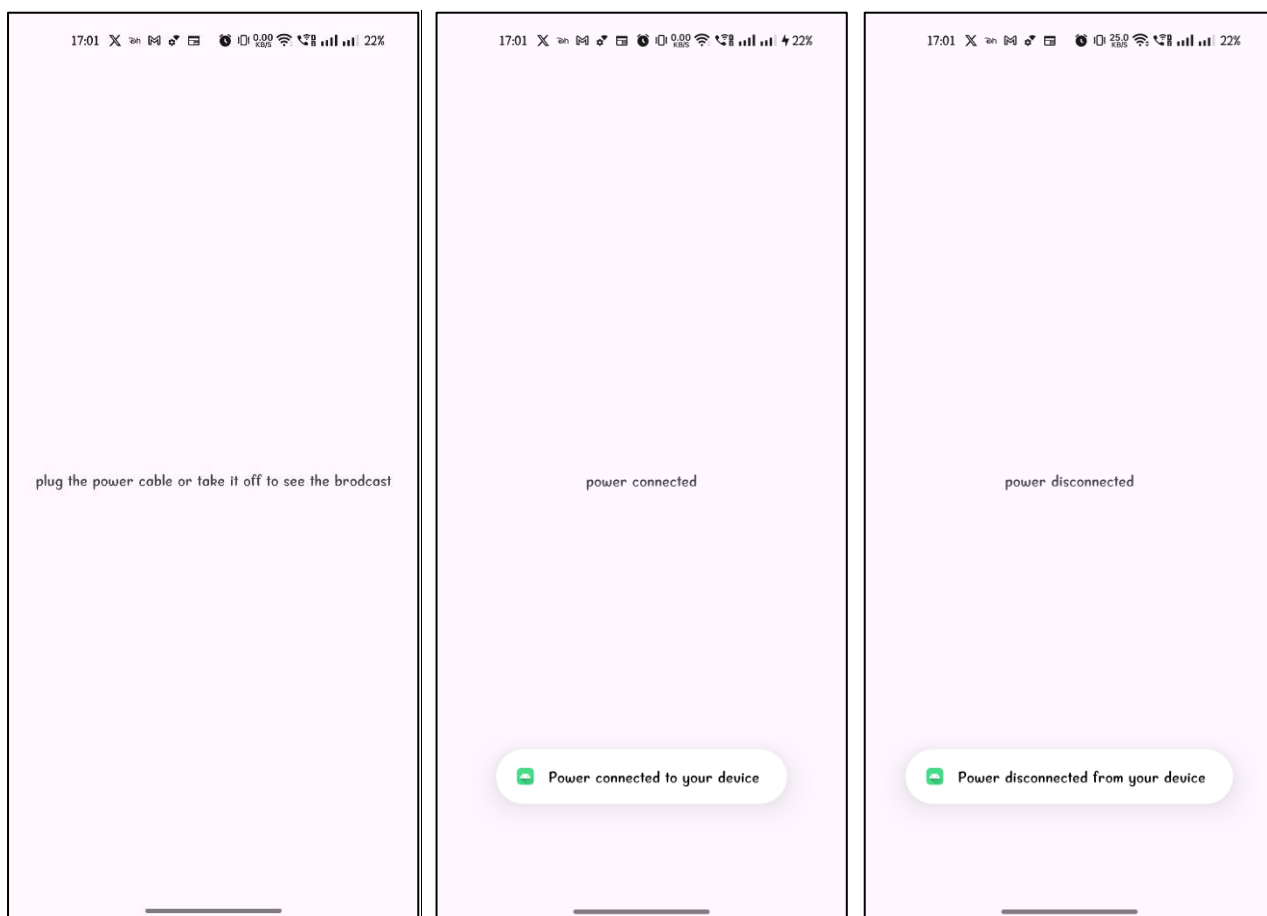
    class Practical12_broadcast extends BroadcastReceiver {
        @Override
        public void onReceive(Context context, Intent intent)
```

```

{
    String action = intent.getAction();

    if (action != null) {
        switch (action) {
            case
"android.intent.action.ACTION_POWER_CONNECTED":
                Toast.makeText(context, "Power connected
to your device", Toast.LENGTH_SHORT).show();
                textView.setText("power connected");
                break;
            case
"android.intent.action.ACTION_POWER_DISCONNECTED
":
                Toast.makeText(context, "Power
disconnected from your device",
Toast.LENGTH_SHORT).show();
                textView.setText("power disconnected");
                break;
            default:
                Toast.makeText(context, "Unknown action:
" + action, Toast.LENGTH_SHORT).show();
                textView.setText("unknown action
triggered");
                break;
        }
    }
}
}
}
}

```



Practical-13

AIM: Develop an Android application that uses Content Providers to share data between different apps and components.

XML:

Manifest file:

Add this to Manifest.xml

```
<provider
    android:name=".Pr13ContentProvider"
    android:authorities="com.mk.madpractical"
    android:enabled="true"
    android:exported="true" />
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical13_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical13_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="we are trying to receive context from the context
        provider"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <Button
        android:id="@+id/Practical13_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"

    app:layout_constraintTop_toBottomOf="@+id/Practical13_textview"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Pr13ContentProvider.java:

```
package com.mk.madpractical;

import android.content.ContentProvider;
import android.content.ContentValues;
import android.database.Cursor;
```

```
import android.database.MatrixCursor;
import android.net.Uri;

public class Pr13ContentProvider extends ContentProvider {

    public Pr13ContentProvider() {
    }

    @Override
    public int delete(Uri uri, String selection, String[] selectionArgs) {
        // Implement this to handle requests to delete one or more rows.
        throw new UnsupportedOperationException("Not yet
        implemented");
    }

    @Override
    public String getType(Uri uri) {
        // at the given URI.
        throw new UnsupportedOperationException("Not yet
        implemented");
    }

    @Override
    public Uri insert(Uri uri, ContentValues values) {
        throw new UnsupportedOperationException("Not yet
        implemented");
    }

    @Override
    public boolean onCreate() {
        return false;
    }

    @Override
    public Cursor query(Uri uri, String[] projection, String selection,
        String[] selectionArgs, String sortOrder) {
        // Create a Cursor with a single column containing the greeting
        message
        MatrixCursor cursor = new MatrixCursor(new
        String[]{"message"});
        cursor.addRow(new String[]{"Hello from the MAD Practical 12
        Content Provider!"});
        return cursor;
    }

    @Override
    public int update(Uri uri, ContentValues values, String selection,
        String[] selectionArgs) {
        throw new UnsupportedOperationException("Not yet
        implemented");
    }
}
```

Practical13.java:

```
package com.mk.madpractical;
import android.content.ContentResolver;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```

import android.net.Uri;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.activity.EdgeToEdge;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical13 extends AppCompatActivity {
    Button btn;
    TextView textView;
    private static final Uri CONTENT_URI =
Uri.parse("content://com.mk.madpractical/hello");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical13);

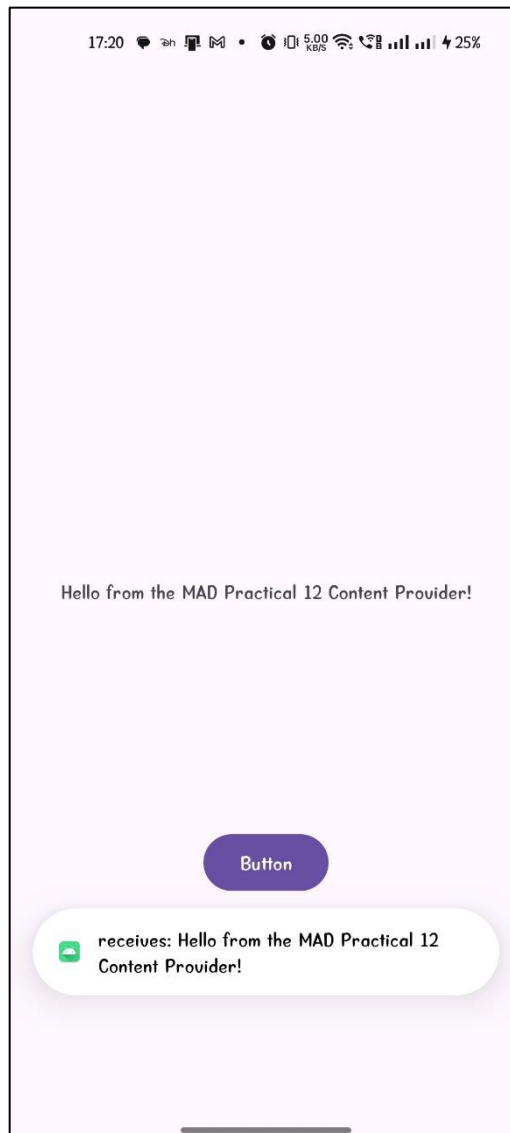
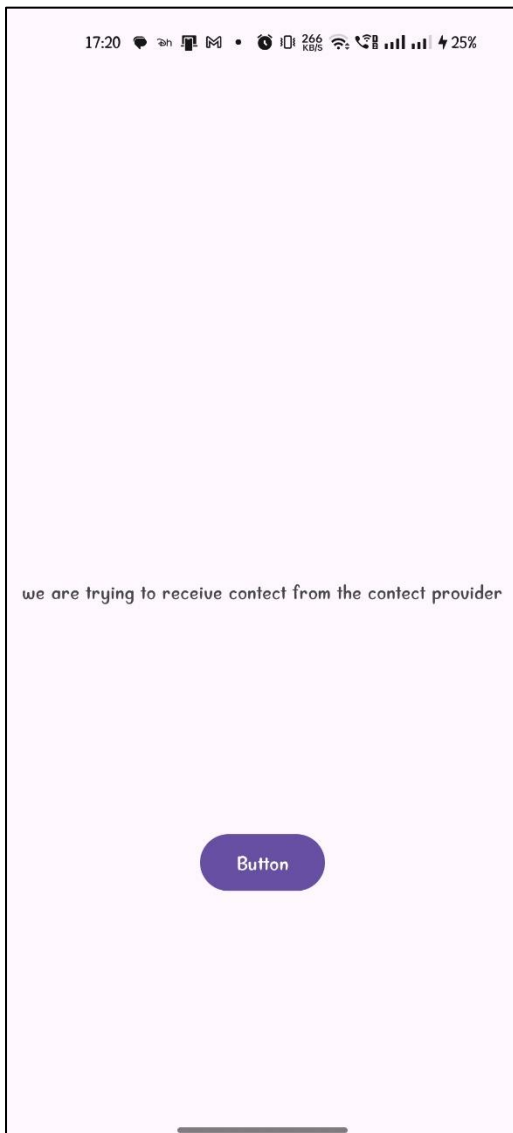
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical13_ConstraintLayout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,

```

```

systemBars.right, systemBars.bottom);
        return insets;
    });
    btn = findViewById(R.id.Practical13_button);
    textView = findViewById(R.id.Practical13_textview);
    btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            ContentResolver contentResolver = getContentResolver();
            Cursor cursor = contentResolver.query(CONTENT_URI, null,
null, null, null);
            if (cursor != null && cursor.moveToFirst()) {
                String msg =
cursor.getString(cursor.getColumnIndex("message"));
                textView.setText(msg);
                Toast.makeText(Practical13.this, "receives: "+msg,
Toast.LENGTH_SHORT).show();
                cursor.close();
            }
        }
    });
}
}

```




```

private ListView contactsListView;
private ArrayList<String> contactsList;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_practical14);

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical14_ConstraintLayout), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
        return insets;
    });

    contactsListView = findViewById(R.id.Practical14_listview);
    contactsList = new ArrayList<>();

    if (ContextCompat.checkSelfPermission(this,
Manifest.permission.READ_CONTACTS)
        != PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.READ_CONTACTS},
PERMISSION_REQUEST_CODE);
    } else {
        loadContacts();
    }
}

private void loadContacts() {
    ContentResolver contentResolver = getContentResolver();
    Uri uri =
android.provider.ContactsContract.CommonDataKinds.Phone.CONTENT_URI;

    Cursor cursor = contentResolver.query(uri, null, null, null, null);

    if (cursor != null) {
        while (cursor.moveToNext()) {
            String name =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.DISPLAY_NAME));
            String phoneNumber =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.NUMBER));
            contactsList.add(name + ": " + phoneNumber);
        }
        cursor.close();
    }

    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, contactsList);
    contactsListView.setAdapter(adapter);
}

@Override
public void onRequestPermissionsResult(int requestCode,

```

```

    } else {
        Toast.makeText(this, "Permission denied to read contacts",
Toast.LENGTH_SHORT).show();
    }
}
}
}
}

```



Practical-15

AIM: Create an application that creates a database using SQLiteOpenHelper Class and performs Insert and Read from the SQLite database.

XML:

View Data activity

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/view_data_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp" >
    <ListView
        android:layout_margin="15dp"
        android:id="@+id/showData"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Practical15 activity

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical15_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical15">
    <EditText
        android:id="@+id/Pr15RollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:ems="10"
        android:hint="Roll Number"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr15Name"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr15Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:ems="10"
```

```
        android:hint="Name"
        android:inputType="text"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr15Semester"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr15Semester"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"
        android:ems="10"
        android:hint="Semester"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
    />
    <Button
        android:id="@+id/Pr15action_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"
        android:text="InsertData"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr15Semester"
    />
    <Button
        android:id="@+id/Pr15view_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="View Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr15action_btn"
    />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

DBHelper.java

```
package com.mk.madpractical;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
```

```

import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.widget.ArrayAdapter;
import java.util.ArrayList;

public class DBHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME =
"StudentData.db";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "StudentRecord";
    private static final String COLUMN_ROLL = "RollNo";
    private static final String COLUMN_NAME = "name";
    private static final String COLUMN_SEM = "Semester";
    private final Context context;

    public DBHelper(Context context) {
        super(context, DATABASE_NAME, null,
DATABASE_VERSION);
        this.context = context;
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE IF NOT EXISTS " +
TABLE_NAME + " (" +
            COLUMN_ROLL + " TEXT PRIMARY KEY," +
            COLUMN_NAME + " TEXT," +
            COLUMN_SEM + " TEXT)";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion,
int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public void insertData(String roll, String name, String sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_ROLL, roll);
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        db.insertWithOnConflict(TABLE_NAME, null,
contentValues, SQLiteDatabase.CONFLICT_REPLACE);
    }

    public Cursor getDataByRoll(String roll) {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME + "
WHERE " + COLUMN_ROLL + "=?";
        Cursor cursor = db.rawQuery(query, new String[]{roll});
        return cursor;
    }

```

```

    public Cursor getAllData() {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME;
        return db.rawQuery(query, null);
    }

    public boolean updateData(String roll, String name, String
sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);
        int result = db.update(TABLE_NAME, contentValues,
COLUMN_ROLL + "=?", new String[]{roll});
        return result > 0; // Return true if update was successful
    }

    // DELETE operation (Delete data by Roll Number)
    public boolean deleteData(String roll) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COLUMN_ROLL +
"=?", new String[]{roll});
        return result > 0; // Return true if delete was successful
    }

    // DELETE operation (Delete all records)
    public void deleteAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        db.execSQL("DELETE FROM " + TABLE_NAME);
    }

    public ArrayAdapter<String> getAdapter(){
        ArrayList<String> rows = new ArrayList<String>();
        Cursor data = getAllData();
        //    data.moveToFirst();
        while(data.moveToNext()){
            String roll =
data.getString(data.getColumnIndex(COLUMN_ROLL));
            String name =
data.getString(data.getColumnIndex(COLUMN_NAME));
            String sem =
data.getString(data.getColumnIndex(COLUMN_SEM));
            String row = "Name: "+name+"\nRoll: "+roll+"\nSem:
"+sem;
            rows.add(row);
        }

        data.close();
        return new ArrayAdapter<String>(context,
androidx.appcompat.R.layout.support_simple_spinner_dropd
own_item, rows);
    }
}

```

ViewData.java

```
package com.mk.madpractical;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;

public class ViewData extends AppCompatActivity {
    private ListView listView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_view_data);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.view_data_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        listView = findViewById(R.id.showData);
        ArrayAdapter adapter = new DBHelper(this).getAdapter();
        if (adapter.getCount() > 0) {
            listView.setAdapter(adapter);
        } else {
            Toast.makeText(this, "The Database is empty!",
            Toast.LENGTH_SHORT).show();
        }
    }
}
```

MainActivity.java

```
package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;

public class Practical15 extends AppCompatActivity {
    TextView rollNo,name,sem;
    Button action,view;
    ListView listView;
    DBHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical15);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical15_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        rollNo = findViewById(R.id.Pr15RollNo);
        name = findViewById(R.id.Pr15Name);
        sem = findViewById(R.id.Pr15Semester);
        action = findViewById(R.id.Pr15action_btn);
        view = findViewById(R.id.Pr15view_btn);
        dbHelper = new DBHelper(this);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                dbHelper.insertData(rollNo.getText().toString(),name.getText().
                toString(),sem.getText().toString());
                Toast.makeText(Practical15.this, "Data Inserted
                Sucessfully", Toast.LENGTH_SHORT).show();
                rollNo.setText("");
                name.setText("");
                sem.setText("");
            }
        });

        view.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new
                Intent(Practical15.this,ViewData.class));
            }
        });
    }
}
```

Output:

14:001 device2.00 Kbps49%75%

123456

studentname

5

InsertData

View Data

12

3

4

5

6

7

8

9

,

0

.

✖

✓

14:011 device21.0 Kbps49%75%

123456

studentname

5

InsertData

View Data

🕒 Data Inserted Sucessfully

14:011 device1.00 Kbps49%75%

Name: moh Roll: 11 Sem: 3

Name: moh Roll: 12 Sem: 3

Name: moh Roll: 13 Sem: 3

Name: moh Roll: 15 Sem: 3

Name: mohit Roll: 16 Sem: 5

Name: moh Roll: 17 Sem: 3

Name: moh Roll: 18 Sem: 3

Name: moh Roll: 19 Sem: 3

Name: studentname Roll: 123456 Sem: 5

Practical-16

AIM: Create an application to Update and Delete data from the SQLite database using SQLiteOpenHelper class

XML:

View Data activity

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/view_data_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp" >
    <ListView
        android:layout_margin="15dp"
        android:id="@+id/showData"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Practical15 activity

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical16_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical16">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:text="Enter the Rollno and new name and semester to update the record and enter the Rollno to delete the record"
        app:layout_constraintBottom_toTopOf="@+id/Pr16RollNo"
        tools:layout_editor_absoluteX="10dp" />
    <EditText
        android:id="@+id/Pr16RollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        android:ems="10"
        android:hint="Roll Number"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr16Name"
        app:layout_constraintEnd_toEndOf="parent"
```

```
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr16Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        android:ems="10"
        android:hint="Name"
        android:inputType="text"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr16Semester"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr16Semester"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:ems="10"
        android:hint="Semester"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        />
    <Button
        android:id="@+id/Pr16action_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Update Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr16Semester"
        />
    <Button
        android:id="@+id/Pr16delete_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="Delete Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr16action_btn"
        />
```

```

<Button
    android:id="@+id/Pr16view_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="View Data"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/Pr16delete_btn"
"/>
</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

DBHelper.java

```

package com.mk.madpractical;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.widget.ArrayAdapter;
import java.util.ArrayList;
public class DBHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME =
"StudentData.db";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "StudentRecord";

    private static final String COLUMN_ROLL = "RollNo";
    private static final String COLUMN_NAME = "name";
    private static final String COLUMN_SEM = "Semester";
    private final Context context;

    public DBHelper(Context context) {
        super(context, DATABASE_NAME, null,
DATABASE_VERSION);
        this.context = context;
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE IF NOT EXISTS " +
TABLE_NAME + " (" +
            COLUMN_ROLL + " TEXT PRIMARY KEY," +
            COLUMN_NAME + " TEXT," +
            COLUMN_SEM + " TEXT)";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion,
int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

```

```

    public void insertData(String roll, String name, String sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_ROLL, roll);
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        db.insertWithOnConflict(TABLE_NAME, null,
contentValues, SQLiteDatabase.CONFLICT_REPLACE);
    }

    public Cursor getDataByRoll(String roll) {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME + "
WHERE " + COLUMN_ROLL + "=?";
        Cursor cursor = db.rawQuery(query, new String[]{roll});
        return cursor;
    }

    public Cursor getAllData() {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME;
        return db.rawQuery(query, null);
    }

    public boolean updateData(String roll, String name, String
sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        int result = db.update(TABLE_NAME, contentValues,
COLUMN_ROLL + "=?", new String[]{roll});
        return result > 0; // Return true if update was successful
    }

    // DELETE operation (Delete data by Roll Number)
    public boolean deleteData(String roll) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COLUMN_ROLL +
"=?", new String[]{roll});
        return result > 0; // Return true if delete was successful
    }

    // DELETE operation (Delete all records)
    public void deleteAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        db.execSQL("DELETE FROM " + TABLE_NAME);
    }

    public ArrayAdapter<String> getAdapter(){
        ArrayList<String> rows = new ArrayList<String>();
        Cursor data = getAllData();
        // data.moveToFirst();
        while(data.moveToNext()){
            String roll =
data.getString(data.getColumnIndex(COLUMN_ROLL));
            String name =
data.getString(data.getColumnIndex(COLUMN_NAME));
            String sem =
data.getString(data.getColumnIndex(COLUMN_SEM));
            String row = "Name: "+name+"\nRoll: "+roll+"\nSem:

```



```

"+sem;
    rows.add(row);
}
data.close();
return new ArrayAdapter<String>(context,
androidx.appcompat.R.layout.support_simple_spinner_dropd
own_item, rows);
}
}

```

ViewData.java

```
package com.mk.madpractical;
```

```

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;

```

```

public class ViewData extends AppCompatActivity {
    private ListView listView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_view_data);

```

```

ViewCompat.setOnApplyWindowInsetsListener(findViewById(
R.id.view_data_layout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
    return insets;
});
listView = findViewById(R.id.showData);
ArrayAdapter adapter = new DBHelper(this).getAdapter();
if (adapter.getCount() > 0) {
    listView.setAdapter(adapter);
} else {
    Toast.makeText(this, "The Database is empty!",
Toast.LENGTH_SHORT).show();
}
}
}

```

MainActivity.java

```
package com.mk.madpractical;
```

```

import android.content.Intent;
import android.os.Bundle;

```

```

import android.view.View;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```

public class Practical16 extends AppCompatActivity {
    TextView rollno, name, sem;
    Button action, view, delete;
    DBHelper dbHelper;

```

```

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical16);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
R.id.Practical16_layout), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

```

```

        rollno = findViewById(R.id.Pr16RollNo);
        name = findViewById(R.id.Pr16Name);
        sem = findViewById(R.id.Pr16Semester);
        action = findViewById(R.id.Pr16action_btn);
        delete = findViewById(R.id.Pr16delete_btn);
        view = findViewById(R.id.Pr16view_btn);

```

```

        dbHelper = new DBHelper(this);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (dbHelper.updateData(rollno.getText().toString(),
name.getText().toString(), sem.getText().toString())) {
                    Toast.makeText(Practical16.this, "Data Updated
Sucessfully", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(Practical16.this, "There is problem
Updating the data may be the RollNo is not found in the
database", Toast.LENGTH_SHORT).show();
                }
            }
        });

```

```

        view.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent(Practical16.this,

```

```

ViewData.class));
    }
    });
    delete.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            if (dbHelper.deleteData(rollno.getText().toString())){
                Toast.makeText(Practical16.this, "Data deleted
                sucessfully", Toast.LENGTH_SHORT).show();
            }else {

```

```

                Toast.makeText(Practical16.this, "There is problem
                Deleting the data may be the RollNo is not found in the
                database", Toast.LENGTH_SHORT).show();
            }
        }
    });
}
}

```

Output:

The output consists of six screenshots arranged in two rows of three, showing the application's interface and its response to user actions.

Top Row:

- Left:** The initial screen with input fields for "Roll Number", "Name", and "Semester". Below the fields are three buttons: "Update Data", "Delete Data", and "View Data".
- Middle:** The "Roll Number" field contains "123456" and the "Name" field contains "update student". The "Semester" field contains "1".
- Right:** The same screen as the middle one, but with a toast message at the bottom: "Data Updated Sucessfully".

Bottom Row:

- Left:** A list of student records displayed in a scrollable view. The records are: "Name: moh Roll: 11 Sem: 3", "Name: moh Roll: 12 Sem: 3", "Name: moh Roll: 13 Sem: 3", "Name: moh Roll: 15 Sem: 3", "Name: mohit Roll: 16 Sem: 5", "Name: moh Roll: 17 Sem: 3", "Name: moh Roll: 18 Sem: 3", "Name: moh Roll: 19 Sem: 3", and "Name: update student Roll: 123456 Sem: 1".
- Middle:** The "Roll Number" field contains "18". The "Name" field is empty, and the "Semester" field contains "1".
- Right:** The same screen as the middle one, but with a toast message at the bottom: "Data deleted sucessfully".

Practical-17

AIM: Perform Firebase Integration to your Android application and store the data in the Firebase Database.

Steps to Perform:

1. Create the project and setup the mainactivity
2. Go to tools -> firebase -> Realtime Database -> Get Started with realtime database java
3. Click on connect it will opens a tab on the browser
4. Now login and create the project (Name of the project will automatically typed)
5. After creating the project click on the connect it will connect the project with firebase project
6. It will create the **google-services.json** in the app directory
7. Now in the firebase console go to build and realtime database
8. Create a realtime database with test mode as security
9. Now the rest of the process is in the code

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical17_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical17">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Enter the values that you want to enter in
the Firebase RealTime Database"

    app:layout_constraintBottom_toTopOf="@+id/Pr17RollNo"
        android:layout_marginBottom="17dp"
        />

    <EditText
        android:id="@+id/Pr17RollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="17dp"
        android:ems="10"
        android:hint="Roll Number"
        android:inputType="numberDecimal"
        android:text=""

    app:layout_constraintBottom_toTopOf="@+id/Pr17Name"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />

    <EditText
        android:id="@+id/Pr17Name"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:layout_marginBottom="17dp"
        android:ems="10"
        android:hint="Name"
        android:inputType="text"
        android:text=""
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
```

```
<EditText
    android:id="@+id/Pr17Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="17dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toBottomOf="@+id/Pr17Name"
/>
```

```
<Button
    android:id="@+id/Pr17action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="17dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toBottomOf="@+id/Pr17Semester"
/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;

import java.util.HashMap;

public class Practical17 extends AppCompatActivity {
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical17);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical17_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        rollno = findViewById(R.id.Pr17RollNo);
        name = findViewById(R.id.Pr17Name);
        sem = findViewById(R.id.Pr17Semester);
        action = findViewById(R.id.Pr17action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
public void onClick(View v) {
    String nameval = name.getText().toString();
    String rollnoval = rollno.getText().toString();
    String semval = sem.getText().toString();

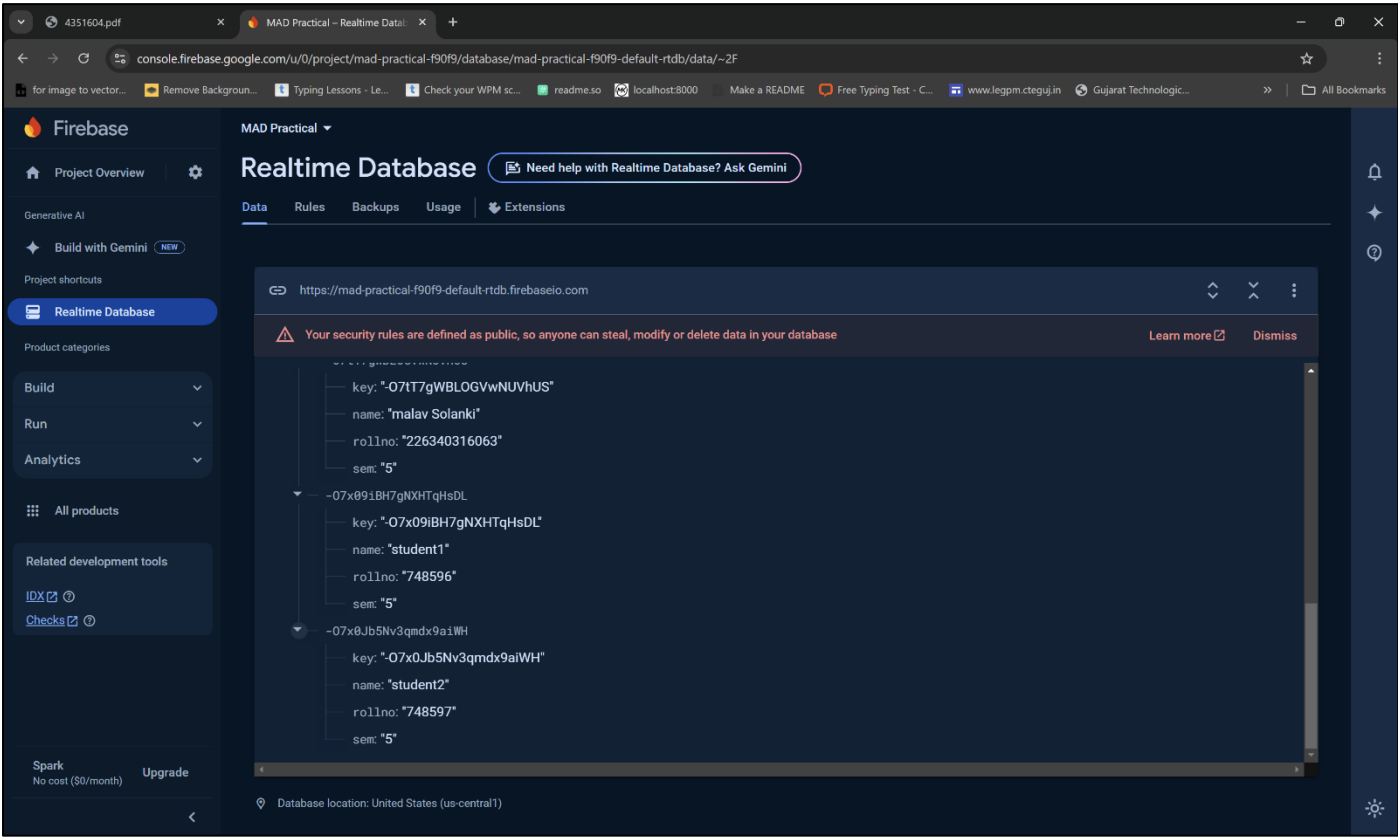
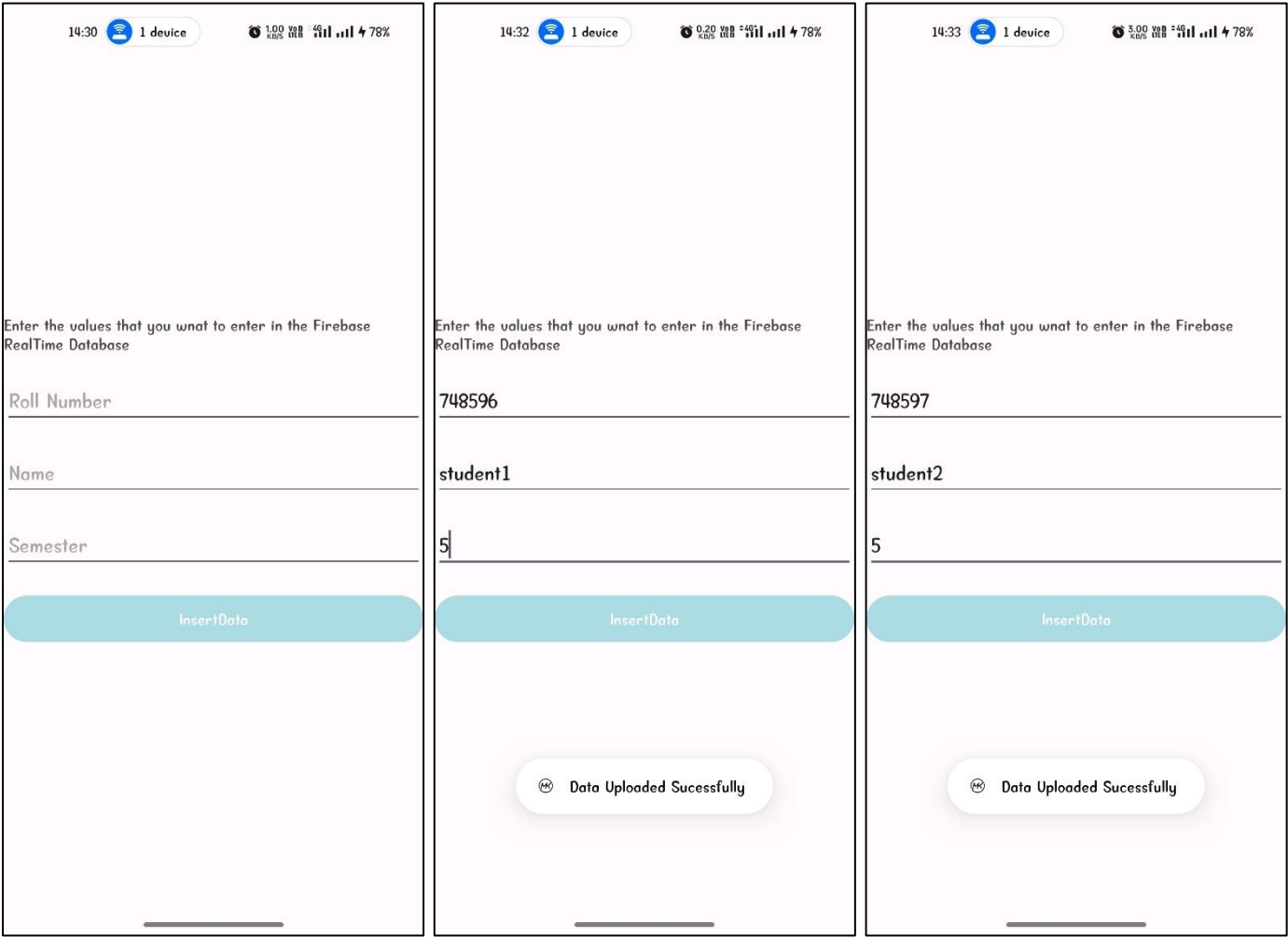
    if (rollnoval.isEmpty()) {
        rollno.setError("RollNo Cannot be empty");
        return;
    }
    if (nameval.isEmpty()) {
        name.setError("Name Cannot be empty");
        return;
    }
    if (semval.isEmpty()) {
        sem.setError("Semester Cannot be empty");
        return;
    }
    addData(rollnoval,nameval,semval);
}
});
}

private void addData(String rollnoval,String nameval,String
semval) {
    HashMap<String , String> values = new HashMap<>();
    values.put("rollno",rollnoval);
    values.put("name",nameval);
    values.put("sem",semval);

    FirebaseDatabase database =
    FirebaseDatabase.getInstance();
    DatabaseReference studentsRef =
    database.getReference("Students");
    String key = studentsRef.push().getKey();
    values.put("key",key);

    studentsRef.child(key).setValue(values).addOnCompleteListen
er(new OnCompleteListener<Void>() {
    @Override
    public void onComplete(@NonNull Task<Void> task) {
        Toast.makeText(Practical17.this, "Data Uploaded
        Sucessfully", Toast.LENGTH_SHORT).show();
        rollno.setText("");
        name.setText("");
        sem.setText("");
    }
});
}
}
```

Output:



Practical-18

AIM: Create an application to retrieve data from the Firebase Database and display it in the RecyclerView

Follow the Steps from Practical-17 to Connect with the database

XML:

View layout for RecyclerView View:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18name"
        android:layout_marginLeft="10dp"
        android:textSize="15sp"
        android:text="Name: Student Name"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18rollno"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_below="@+id/Pr18name"
        android:text="RollNo: 226340316099"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18sem"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_marginBottom="10dp"
        android:layout_below="@+id/Pr18rollno"
        android:text="Semester: 5"/>
    <View
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:background="?android:attr/textColorSecondary"
        android:layout_below="@+id/Pr18sem" />
</RelativeLayout>
```

MainActivity:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical18layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical18">
```

```
<androidx.recyclerview.widget.RecyclerView
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/Pr18Recyclerview"/>
```

```
</RelativeLayout>
```

JAVA:

Student Class (DataHolder):

```
package com.mk.madpractical;
```

```
import java.util.Objects;
```

```
public class Student {
    String key; // Firebase key
    String rollno;
    String name;
    String sem;

    public Student(String key, String rollno, String name, String
sem) {
        this.key = key;
        this.rollno = rollno;
        this.name = name;
        this.sem = sem;
    }

    public Student() {
        // Default constructor required for calls to
        dataSnapshot.getValue(Student.class)
    }

    // Getter and setter for key
    public String getKey() {
        return key;
    }

    public void setKey(String key) {
        this.key = key;
    }

    public String getRollno() {
        return rollno;
    }

    public void setRollno(String rollno) {
        this.rollno = rollno;
    }
}
```

```

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public String getSem() {
    return sem;
}

public void setSem(String sem) {
    this.sem = sem;
}
}

```

StudentHolder (ViewHolder for RecyclerView):

```

package com.mk.madpractical;
import android.view.View;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class StudentHolder extends RecyclerView.ViewHolder {
    TextView name,rollno,sem;
    public StudentHolder(@NonNull View itemView) {
        super(itemView);
        name = itemView.findViewById(R.id.Pr18name);
        rollno = itemView.findViewById(R.id.Pr18rollno);
        sem = itemView.findViewById(R.id.Pr18sem);
    }
}

```

StudentAdapter (Adapter Class for RecyclerView):

```

package com.mk.madpractical;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class StudentAdapter extends
RecyclerView.Adapter<StudentHolder> {
    public StudentAdapter(Context context, List<Student>
students) {
        this.context = context;
        this.students = students;
    }

    Context context;
    List<Student> students;
    @NonNull

```

```

@Override
    public StudentHolder onCreateViewHolder(@NonNull
ViewGroup parent, int viewType) {
        return new
StudentHolder(LayoutInflater.from(context).inflate(R.layout.pr
18recyclerview,parent,false));
    }

    @Override
    public void onBindViewHolder(@NonNull StudentHolder
holder, int position) {
        Student student = students.get(position);
        holder.rollno.setText("RollNO: "+student.rollno);
        holder.name.setText("Name: "+student.name);
        holder.sem.setText("Semester: "+student.sem);
    }

    @Override
    public int getItemCount() {
        return students.size();
    }

    public void addStudent(Student student) {
        if (getStudentPositionByKey(student.key) == -1) {
            students.add(student);
            notifyItemInserted(students.size() - 1);
        }
    }

    public void updateStudent(Student updatedStudent) {
        int position =
getStudentPositionByKey(updatedStudent.key);
        if (position != -1) {
            students.set(position, updatedStudent);
            notifyItemChanged(position);
        }
    }

    public void removeStudent(String key) {
        int position = getStudentPositionByKey(key);
        if (position != -1) {
            students.remove(position);
            notifyItemRemoved(position);
        }
    }

    private int getStudentPositionByKey(String key) {
        for (int i = 0; i < students.size(); i++) {
            if (students.get(i).getKey().equals(key)) {
                return i;
            }
        }
        return -1;
    }
}

```


Main Activity:

```
package com.mk.madpractical;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import com.google.firebase.Firebase;
import com.google.firebase.database.ChildEventListener;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;
import java.util.ArrayList;
import java.util.List;

public class Practical18 extends AppCompatActivity {
    RecyclerView recyclerView;

    private ArrayList<Student> students;
    private StudentAdapter adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical18);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical18layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        recyclerView = findViewById(R.id.Pr18Recyclerview);
        recyclerView.setLayoutManager(new
        LinearLayoutManager(this));
        students = new ArrayList<Student>();
        adapter = new StudentAdapter(this, students);
        recyclerView.setAdapter(adapter);
        DatabaseReference databaseReference =
        FirebaseDatabase.getInstance().getReference("Students");

        databaseReference.addChildEventListener(new
```

```
ChildEventListener() {

        @Override
        public void onChildAdded(@NonNull DataSnapshot
        dataSnapshot, String previousChildName) {
            // Called when a new student is added to the
            database
            Student student =
            dataSnapshot.getValue(Student.class);
            if (student != null) {
                student.setKey(dataSnapshot.getKey());
                adapter.addStudent(student);
            }
        }

        @Override
        public void onChildChanged(DataSnapshot
        dataSnapshot, String previousChildName) {
            // Called when a student is updated in the database
            String key = dataSnapshot.getKey();
            Student updatedStudent =
            dataSnapshot.getValue(Student.class);
            if (updatedStudent != null) {
                updatedStudent.setKey(key); // Set the key
                adapter.updateStudent(updatedStudent);
            }
        }

        @Override
        public void onChildRemoved(DataSnapshot
        dataSnapshot) {
            // Called when a student is removed from the
            database
            String key = dataSnapshot.getKey();
            adapter.removeStudent(key);
        }

        @Override
        public void onChildMoved(DataSnapshot dataSnapshot,
        String previousChildName) {
            // Not used in this case
        }

        @Override
        public void onCancelled(DatabaseError databaseError) {
            // Handle possible errors
        }
    }
}
```

Output:

14:59

 1 device

 0.88 KB/S

 4G



 79%

Name: Mohit Makwana

RollNO: 226340316036

Semester: 5

Name: Makwana Mohit

RollNO: 226340316036

Semester: 5

Name: Neha Kavaiya

RollNO: 226340316029

Semester: 5

Name: Mohit Makwana

RollNO: 235

Semester: 6

Name: malav Solanki

RollNO: 226340316063

Semester: 5

Name: student1

RollNO: 748596

Semester: 5

Name: student2

RollNO: 748597

Semester: 5

14:59

 1 device

 5.00 KB/S

 4G



 79%

Name: MONIT MARWANA

RollNO: 226340316036

Semester: 5

Name: Makwana Mohit

RollNO: 226340316036

Semester: 5

Name: Neha Kavaiya

RollNO: 226340316029

Semester: 5

Name: Mohit Makwana

RollNO: 235

Semester: 6

Name: malav Solanki

RollNO: 226340316063

Semester: 5

Name: student1

RollNO: 748596

Semester: 5

Name: student2

RollNO: 748597

Semester: 5

Practical-19

AIM: Connect an Android application to the MySQL database using PHP, and insert the data in the database table.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr19.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```
<?php
$servername = "localhost"; // Database server
$username = "mad"; // Database username
$password = "hellomad"; // Database password
$dbname = "maD"; // Database name

// Create connection
$conn = new mysqli($servername, $username, $password,
$dbname);
if (empty($_POST)){
    die("post method required");
}
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Get POST data from Android app
$rollno = $_POST['rollno'];
$name = $_POST['name'];
$sem = $_POST['sem'];
$sql = "INSERT INTO Student (rollno, name, sem) VALUES
('$rollno', '$name', '$sem)";

if ($conn->query($sql) === TRUE) {
    echo "Data inserted successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical19_layout"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp"
tools:context=".Practical19">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the values that you want to enter in
the Firebase RealTime Database"
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr19RollNo"
    android:layout_marginBottom="15dp"
/>
```

```
<EditText
    android:id="@+id/Pr19RollNo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Roll Number"
    android:inputType="numberDecimal"
    android:text=""
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr19Name"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<EditText
    android:id="@+id/Pr19Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
```

```

app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr19Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr19Name"
/>

<Button
    android:id="@+id/Pr19action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr19Semester"
/>

</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

```

package com.mk.madpractical;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.OutputStream;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.HashMap;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

```

```

public class Practical19 extends AppCompatActivity {
    private ExecutorService executorService;
    private Handler mainHandler;
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical19);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical19_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        executorService = Executors.newSingleThreadExecutor();
        mainHandler = new Handler(Looper.getMainLooper());
        rollno = findViewById(R.id.Pr19RollNo);
        name = findViewById(R.id.Pr19Name);
        sem = findViewById(R.id.Pr19Semester);
        action = findViewById(R.id.Pr19action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String nameval = name.getText().toString();
                String rollnoval = rollno.getText().toString();
                String semval = sem.getText().toString();

                if (rollnoval.isEmpty()) {
                    rollno.setError("RollNo Cannot be empty");
                    return;
                }
                if (nameval.isEmpty()) {
                    name.setError("Name Cannot be empty");
                    return;
                }
                if (semval.isEmpty()) {
                    sem.setError("Semester Cannot be empty");
                    return;
                }
                insertData(rollnoval, nameval, semval);
            }
        });

        private void insertData(final String rollnoval, final String
            nameval, final String semval) {
            executorService.execute(new Runnable() {
                @Override
                public void run() {

```

```

try {
    // Make the HTTP request in the background thread
    URL url = new
URL("http://192.168.79.127:5700/MAD-Pr19.php");
    HttpURLConnection conn = (HttpURLConnection)
url.openConnection();
    conn.setRequestMethod("POST");
    conn.setDoOutput(true);
    conn.setDoInput(true);
    conn.setRequestProperty("Content-Type",
"application/x-www-form-urlencoded");

    // Prepare POST data
    String postData = "rollno=" + rollnoval + "&name="
+ nameval + "&sem=" + semval;

    // Send POST data
    OutputStream os = conn.getOutputStream();
    os.write(postData.getBytes());
    os.flush();
    os.close();

    // Get the response from the server
    BufferedReader in = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
    String inputLine;
    StringBuilder response = new StringBuilder();

    while ((inputLine = in.readLine()) != null) {
        response.append(inputLine);
    }
    in.close();

```

```

// Send the server response to the main thread
final String result = response.toString();
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        // Show the server response as a Toast on the
main thread
        Toast.makeText(getApplicationContext(),
result, Toast.LENGTH_LONG).show();
        rollno.setText("");
        name.setText("");
        sem.setText("");
    }
});

} catch (Exception e) {
    Log.e("Error", e.getMessage());

    // Post the error message to the main thread
mainHandler.post(new Runnable() {
        @Override
        public void run() {
            Toast.makeText(getApplicationContext(), "Error
occurred:" + e.getMessage(), Toast.LENGTH_LONG).show();
        }
    });
}
}
}
}
}

```

Output:

15:32 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

Roll Number

Name

Semester

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748595

student1

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748596

student2

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748597

student3

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748598

student4

5

InsertData

15:38 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748599

student5

5

InsertData

```
mysql>
mysql> select * from Student;
+-----+-----+-----+
| rollno | name   | sem  |
+-----+-----+-----+
| 748595 | student1 | 5    |
| 748596 | student2 | 5    |
| 748597 | student3 | 5    |
| 748598 | student4 | 5    |
| 748599 | student5 | 5    |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

Practical-20

AIM: Perform insertion of data to MySQL database using PHP from an Android application.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr20.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```
<?php
$servername = "localhost"; // Database server
$username = "mad"; // Database username
$password = "hellomad"; // Database password
$dbname = "maD"; // Database name

// Create connection
$conn = new mysqli($servername, $username, $password,
$dbname);
if (empty($_POST)){
    die("post method required");
}
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Get POST data from Android app
$rollno = $_POST['rollno'];
$name = $_POST['name'];
$sem = $_POST['sem'];
$sql = "INSERT INTO Student (rollno, name, sem) VALUES
('$rollno', '$name', '$sem)";

if ($conn->query($sql) === TRUE) {
    echo "Data inserted successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical20_layout"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical20">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the values that you want to enter in
the Firebase RealTime Database"
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr20RollNo"
    android:layout_marginBottom="15dp"
/>
```

```
<EditText
    android:id="@+id/Pr20RollNo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Roll Number"
    android:inputType="numberDecimal"
    android:text=""
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr20Name"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<EditText
    android:id="@+id/Pr20Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
```

```

app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr20Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr20Name"
/>

<Button
    android:id="@+id/Pr20action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr20Semester"
/>

</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

```

package com.mk.madpractical;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.OutputStream;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.HashMap;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

```

```

public class Practical19 extends AppCompatActivity {
    private ExecutorService executorService;
    private Handler mainHandler;
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical20);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical19_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        executorService = Executors.newSingleThreadExecutor();
        mainHandler = new Handler(Looper.getMainLooper());
        rollno = findViewById(R.id.Pr20RollNo);
        name = findViewById(R.id.Pr20Name);
        sem = findViewById(R.id.Pr20Semester);
        action = findViewById(R.id.Pr20action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String nameval = name.getText().toString();
                String rollnoval = rollno.getText().toString();
                String semval = sem.getText().toString();

                if (rollnoval.isEmpty()) {
                    rollno.setError("RollNo Cannot be empty");
                    return;
                }
                if (nameval.isEmpty()) {
                    name.setError("Name Cannot be empty");
                    return;
                }
                if (semval.isEmpty()) {
                    sem.setError("Semester Cannot be empty");
                    return;
                }
                insertData(rollnoval, nameval, semval);
            }
        });

        private void insertData(final String rollnoval, final String
            nameval, final String semval) {
            executorService.execute(new Runnable() {
                @Override
                public void run() {

```



```

try {
    // Make the HTTP request in the background thread
    URL url = new
URL("http://192.168.79.127:5700/MAD-Pr20.php");
    HttpURLConnection conn = (HttpURLConnection)
url.openConnection();
    conn.setRequestMethod("POST");
    conn.setDoOutput(true);
    conn.setDoInput(true);
    conn.setRequestProperty("Content-Type",
"application/x-www-form-urlencoded");

    // Prepare POST data
    String postData = "rollno=" + rollnoval + "&name="
+ nameval + "&sem=" + semval;

    // Send POST data
    OutputStream os = conn.getOutputStream();
    os.write(postData.getBytes());
    os.flush();
    os.close();

    // Get the response from the server
    BufferedReader in = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
    String inputLine;
    StringBuilder response = new StringBuilder();

    while ((inputLine = in.readLine()) != null) {
        response.append(inputLine);
    }
    in.close();

```

```

// Send the server response to the main thread
final String result = response.toString();
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        // Show the server response as a Toast on the
main thread
        Toast.makeText(getApplicationContext(),
result, Toast.LENGTH_LONG).show();
        rollno.setText("");
        name.setText("");
        sem.setText("");
    }
});

} catch (Exception e) {
    Log.e("Error", e.getMessage());

    // Post the error message to the main thread
mainHandler.post(new Runnable() {
        @Override
        public void run() {
            Toast.makeText(getApplicationContext(), "Error
occurred:" + e.getMessage(), Toast.LENGTH_LONG).show();
        }
    });
}
}
}
}

```

Output

15:32 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

Roll Number

Name

Semester

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748595

student1

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748596

student2

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748597

student3

5

InsertData

15:34 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748598

student4

5

InsertData

15:38 1 device 82%

Enter the values that you want to enter in the Firebase RealTime Database

748599

student5

5

InsertData

```
mysql>
mysql> select * from Student;
+-----+-----+-----+
| rollno | name   | sem  |
+-----+-----+-----+
| 748595 | student1 | 5    |
| 748596 | student2 | 5    |
| 748597 | student3 | 5    |
| 748598 | student4 | 5    |
| 748599 | student5 | 5    |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

Practical-21

AIM: Perform reading of the data from the MySQL database using PHP in the JSON format and display on the screen of an Android application.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr20.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```
<?php

// Set content type to JSON

header('Content-Type: application/json');

// Database connection details

$host = 'localhost'; // or your database host

$db = 'mad'; // replace with your database name

$user = 'mad'; // replace with your database username

$pass = 'hellomad'; // replace with your database password

// Create connection

$conn = new mysqli($host, $user, $pass, $db);

// Check connection

if ($conn->connect_error) {

    die(json_encode(["error" => "Connection failed: " . $conn->connect_error]));

}

// SQL query to fetch all records from the student table

$sql = "SELECT rollno, name, sem FROM student";

$result = $conn->query($sql);

// Check if records are found

if ($result->num_rows > 0) {

    // Create an array to store the records

    $students = array();

    // Fetch each row and add to the array

    while($row = $result->fetch_assoc()) {

        $students[] = $row;

    }

}
```

```
// Return the data as JSON
```

```
echo json_encode($students);
```

```
} else {
```

```
    // Return an empty array if no records found
```

```
    echo json_encode([]);
```

```
}
```

```
// Close connection
```

```
$conn->close();
```

```
?>
```

XML:

View layout for Recycler View:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18name"
        android:layout_marginLeft="10dp"
        android:textSize="15sp"
        android:text="Name: Student Name"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18rollno"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_below="@+id/Pr18name"
        android:text="RollNo: 226340316099"/>
```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/Pr18sem"
    android:layout_marginTop="10dp"
    android:layout_marginLeft="10dp"
    android:layout_marginBottom="10dp"
    android:layout_below="@+id/Pr18rollno"
    android:text="Semester: 5"/>
<View
    android:layout_width="match_parent"
    android:layout_height="1dp"
    android:background="?android:attr/textColorSecondary"
    android:layout_below="@+id/Pr18sem" />
</RelativeLayout>

```

MainActivity:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    "
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical21layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical21">

    <androidx.recyclerview.widget.RecyclerView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/Pr21Recyclerview"/>

</RelativeLayout>

```

JAVA:

Student Class (DataHolder):

```

package com.mk.madpractical;

import java.util.Objects;

public class Student {
    String key; // Firebase key
    String rollno;
    String name;
    String sem;

    public Student(String key, String rollno, String name, String sem) {
        this.key = key;
        this.rollno = rollno;
        this.name = name;
        this.sem = sem;
    }

    public Student() {
        // Default constructor required for calls to

```

```

    dataSnapshot.getValue(Student.class)
    }

    // Getter and setter for key
    public String getKey() {
        return key;
    }

    public void setKey(String key) {
        this.key = key;
    }

    public String getRollno() {
        return rollno;
    }

    public void setRollno(String rollno) {
        this.rollno = rollno;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getSem() {
        return sem;
    }

    public void setSem(String sem) {
        this.sem = sem;
    }
}

```

StudentHolder (ViewHolder for RecyclerView):

```

package com.mk.madpractical;
import android.view.View;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class StudentHolder extends RecyclerView.ViewHolder {
    TextView name,rollno,sem;
    public StudentHolder(@NonNull View itemView) {
        super(itemView);
        name = itemView.findViewById(R.id.Pr18name);
        rollno = itemView.findViewById(R.id.Pr18rollno);
        sem = itemView.findViewById(R.id.Pr18sem);
    }
}

```

StudentAdapter (Adapter Class for RecyclerView):

```

package com.mk.madpractical;

```

```

import android.content.Context;
import android.view.LayoutInflater;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class StudentAdapter extends
RecyclerView.Adapter<StudentHolder> {
    public StudentAdapter(Context context, List<Student>
students) {
        this.context = context;
        this.students = students;
    }

    Context context;
    List<Student> students;
    @NonNull
    @Override
    public StudentHolder onCreateViewHolder(@NonNull
ViewGroup parent, int viewType) {
        return new
StudentHolder(LayoutInflater.from(context).inflate(R.layout.pr
18recyclerview,parent,false));
    }

    @Override
    public void onBindViewHolder(@NonNull StudentHolder
holder, int position) {
        Student student = students.get(position);
        holder.rollno.setText("RollNO: "+student.rollno);
        holder.name.setText("Name: "+student.name);
        holder.sem.setText("Semester: "+student.sem);
    }

    @Override
    public int getItemCount() {
        return students.size();
    }

    public void addStudent(Student student) {
        if (getStudentPositionByKey(student.key) == -1) {
            students.add(student);
            notifyItemInserted(students.size() - 1);
        }
    }

    public void updateStudent(Student updatedStudent) {
        int position =
getStudentPositionByKey(updatedStudent.key);
        if (position != -1) {
            students.set(position, updatedStudent);
            notifyItemChanged(position);
        }
    }

    public void removeStudent(String key) {
        int position = getStudentPositionByKey(key);

```

```

        if (position != -1) {
            students.remove(position);
            notifyItemRemoved(position);
        }
    }

    private int getStudentPositionByKey(String key) {
        for (int i = 0; i < students.size(); i++) {
            if (students.get(i).getKey().equals(key)) {
                return i;
            }
        }
        return -1;
    }
}

```

Main Activity:

```

package com.mk.madpractical;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import android.util.Log;
import android.widget.Toast;
import org.json.JSONArray;
import org.json.JSONException;
import org.json.JSONObject;
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;
import okhttp3.Call;
import okhttp3.Callback;
import okhttp3.OkHttpClient;
import okhttp3.Request;
import okhttp3.Response;
import java.util.ArrayList;

public class Practical21 extends AppCompatActivity {
    RecyclerView recyclerView;
    private ArrayList<Student> students;
    private StudentAdapter adapter;
    private static final String URL_STRING =
"http://192.168.66.127:5700/MAD-Pr21.php"; // Replace with
your PHP URL
    private final OkHttpClient client = new OkHttpClient();
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical21);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R
.id.Practical21layout), (v, insets) -> {
            Insets systemBars =

```

```

insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
    return insets;
});
recyclerView = findViewById(R.id.Pr21Recyclerview);
recyclerView.setLayoutManager(new
LinearLayoutManager(this));
students = new ArrayList<Student>();
adapter = new StudentAdapter(this, students);
recyclerView.setAdapter(adapter);
fetchStudentData();
}
private void fetchStudentData() {
    // Build the request to fetch data from the server
    Request request = new Request.Builder()
        .url(URL_STRING)
        .build();
    // Make an asynchronous call using OkHttp
    client.newCall(request).enqueue(new Callback() {
        @Override
        public void onFailure(Call call, IOException e) {
            Toast.makeText(Practical21.this, "Error fetching the
data", Toast.LENGTH_SHORT).show();
        }
        @Override
        public void onResponse(Call call, Response response)
throws IOException {
            if (response.isSuccessful()) {
                final String responseData =
response.body().string();
                runOnUiThread(() -> {
                    try {
                        parseJsonToStudentList(responseData);
                    } catch (JSONException e) {

```

```

                        Log.e("MainActivity", "Error parsing JSON", e);
                    }
                });
            }}
        });
    }

    // Method to parse the JSON response and return a list of
Student objects
    private List<Student> parseJsonToStudentList(String
jsonData) throws JSONException {
        List<Student> studentList = new ArrayList<>();
        // Parse the JSON array from the response
        JSONArray jsonArray = new JSONArray(jsonData);

        // Loop through each object in the JSON array
        for (int i = 0; i < jsonArray.length(); i++) {
            JSONObject studentJson = jsonArray.getJSONObject(i);

            // Extract the values and create a Student object
            String rollno = studentJson.getString("rollno");
            String name = studentJson.getString("name");
            String sem = studentJson.getString("sem");

            // Create the Student object, keeping key same as rollno
            Student student = new Student(rollno, rollno, name,
sem);

            // Add the student object to the list
            adapter.addStudent(student);
            studentList.add(student);
        }

        return studentList;

```

Output:

16:021 device5.00 KB/s4G44% Name: studenta RollNO: 123789 Semester: 5 Name: Neha Kavaiya RollNO: 2263403160 Semester: 5 Name: student1 RollNO: 748595 Semester: 5 Name: student2 RollNO: 748596 Semester: 5 Name: student3 RollNO: 748597 Semester: 5 Name: student4 RollNO: 748598 Semester: 5 Name: student5 RollNO: 748599	16:021 device0.13 KB/s4G44% Name: hello RollNO: 123456789 Semester: 7 Name: hello2 RollNO: 123456789123 Semester: 6 Name: studentb RollNO: 123781 Semester: 5 Name: studentc RollNO: 123782 Semester: 5 Name: studentd RollNO: 123783 Semester: 5 Name: studente RollNO: 123784 Semester: 5 Name: studentf RollNO: 123785 Semester: 5
---	---

Practical-22

AIM: Integrate Google maps API to your Android application and display your current location in the app.

Steps to follow:

1. Create a new project
2. Then right click on app > new > google > google maps activity
3. In Manifest file you see a todo and in paste your api key in the api key section
4. You can use below api key if you don't have any
5. <meta-data
 android:name="com.google.android.geo.API_KEY"
 android:value="AlzaSyDV2_xy58r15K6TskZy4KWMuhUDVq67jqM" />

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<fragment
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:map="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/map"

    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical22" />
```

JAVA:

```
package com.mk.madpractical;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.fragment.app.FragmentActivity;
import android.content.Context;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Toast;
import com.google.android.gms.maps.model.Marker;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.mk.madpractical.databinding.ActivityPractical22Binding;

public class Practical22 extends FragmentActivity implements
    OnMapReadyCallback {
    private static final int LOCATION_PERMISSION_REQUEST_CODE =
    1;

    private GoogleMap mMap;
    private LocationManager locationManager;
    private ActivityPractical22Binding binding;
```

```
private ActivityResultLauncher<Intent> locationLauncher;
private Marker currentMarker;
private final LocationListener locationListener = new
    LocationListener() {
    @Override
    public void onLocationChanged(@NonNull Location location) {

        LatLng newLocation = new
        LatLng(location.getLatitude(),location.getLongitude());
        currentMarker.setPosition(newLocation);
        // Optionally, move the camera to the new position

        mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(newLo
        cation, 15));

    }
    @Override
    public void onStatusChanged(String provider, int status, Bundle
    extras) {
    }
    @Override
    public void onProviderEnabled(@NonNull String provider) {
    }
    @Override
    public void onProviderDisabled(@NonNull String provider) {
        Toast.makeText(getApplicationContext(), "Please enable GPS
        and Internet", Toast.LENGTH_LONG).show();
    }
};

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    binding = ActivityPractical22Binding.inflate(getLayoutInflater());
    setContentView(binding.getRoot());
    // Obtain the SupportMapFragment and get notified when the
    map is ready to be used.
    SupportMapFragment mapFragment = (SupportMapFragment)
    getSupportFragmentManager()
        .findFragmentById(R.id.map);
    mapFragment.getMapAsync(this);
    locationManager = (LocationManager)
    getSystemService(Context.LOCATION_SERVICE);
}

@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;
```

```

        addPointer();
    }
    private void addPointer() {
        if (ContextCompat.checkSelfPermission(this,
        android.Manifest.permission.ACCESS_FINE_LOCATION) !=
        PackageManager.PERMISSION_GRANTED &&
        ContextCompat.checkSelfPermission(this,
        android.Manifest.permission.ACCESS_COARSE_LOCATION) !=
        PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, new
            String[]{android.Manifest.permission.ACCESS_FINE_LOCATION,
            android.Manifest.permission.ACCESS_COARSE_LOCATION},
            LOCATION_PERMISSION_REQUEST_CODE);
        } else {
            Location location =
            locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            LatLng initialLocation = new
            LatLng(location.getLatitude(),location.getLongitude());
            currentMarker = mMap.addMarker(new
            MarkerOptions().position(initialLocation).title("Initial Location"));

            mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(initialLocation, 15));

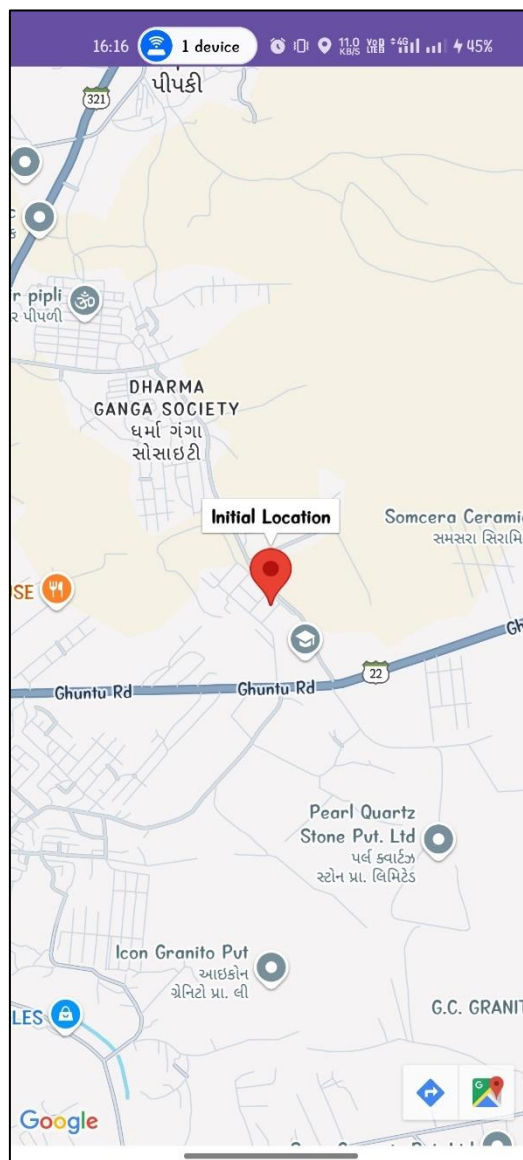
```

```

        locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER, 1, 1, locationListener);
    }
    @Override
    public void onRequestPermissionsResult(int requestCode,
    @NonNull String[] permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions,
        grantResults);
        if (requestCode == LOCATION_PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] ==
            PackageManager.PERMISSION_GRANTED) {
                // Obtain the SupportMapFragment and get notified when
                the map is ready to be used.
                SupportMapFragment mapFragment =
                (SupportMapFragment) getSupportFragmentManager()
                .findFragmentById(R.id.map);
                mapFragment.getMapAsync(this);
            }else{
                Toast.makeText(this, "location permission required to add
                pointer", Toast.LENGTH_SHORT).show();
            }
        }
    }
}

```

Output:



Practical-23

AIM: Integrate Google maps API to your Android application and display your current location in the app.

Steps to follow:

1. Create a new project
2. Then right click on app > new > google > google maps activity
3. In Manifest file you see a todo and in paste your api key in the api key section
4. You can use below api key if you don't have any
5.

```
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value="AlzaSyDV2_xy58r15K6TskZy4KWMuhUDVq67jqM" />
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<fragment
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:map="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/map"

    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical23" />
```

JAVA:

```
package com.mk.madpractical;
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.fragment.app.FragmentActivity;
import android.content.Context;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Toast;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.mk.madpractical.databinding.ActivityPractical23Binding;
import com.google.android.gms.maps.model.Marker;

public class Practical23 extends FragmentActivity implements
    OnMapReadyCallback {

    private GoogleMap mMap;
    private ActivityPractical23Binding binding;
    private final double Rad = 6371000;
    private LocationManager locationManager;
    private double lastLat, lastLng;
    private static final int LOCATION_PERMISSION_REQUEST_CODE =
    1;
    private Marker newMarker;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    binding = ActivityPractical23Binding.inflate(getLayoutInflater());
    setContentView(binding.getRoot());
    locationManager = (LocationManager)
    getSystemService(Context.LOCATION_SERVICE);

    getLocation();
}
@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;
    LatLng initialLocation = new LatLng(lastLat, lastLng);
    mMap.addMarker(new
    MarkerOptions().position(initialLocation).title("Your Last Location"));

    mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(initialL
    ocation, 15));
    newMarker = mMap.addMarker(new
    MarkerOptions().position(initialLocation).title("Your new Location"));
    mMap.setOnMapClickListener(latLng -> {
        newMarker.setPosition(latLng);

        double distance =
        calculateDistance(latLng.latitude, latLng.longitude, lastLat, lastLng);
        Toast.makeText(this, "Distance between pointers is:
        "+String.format("%.2f", distance/1000)+"Km",
        Toast.LENGTH_SHORT).show();
    });
}
private double calculateDistance(double lat1, double lon1, double
lat2, double lon2) {
    // Convert latitude and longitude from degrees to radians
    double lat1Rad = Math.toRadians(lat1);
    double lat2Rad = Math.toRadians(lat2);
    double deltaLat = Math.toRadians(lat2 - lat1);
    double deltaLon = Math.toRadians(lon2 - lon1);

    // Haversine formula
    double a = Math.sin(deltaLat / 2) * Math.sin(deltaLat / 2) +
        Math.cos(lat1Rad) * Math.cos(lat2Rad) *
        Math.sin(deltaLon / 2) * Math.sin(deltaLon / 2);
    double c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
```

```

// Distance in meters
return Rad * c;
}
@Override
public void onRequestPermissionsResult(int requestCode,
@NonNull String[] permissions, @NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions,
grantResults);
    if (requestCode == LOCATION_PERMISSION_REQUEST_CODE) {
        if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            getLocation();
        } else {
            Toast.makeText(this, "location permission required to add
pointer", Toast.LENGTH_SHORT).show();
        }
    }
}

private void getLocation() {
    if (ContextCompat.checkSelfPermission(this,
android.Manifest.permission.ACCESS_FINE_LOCATION) !=

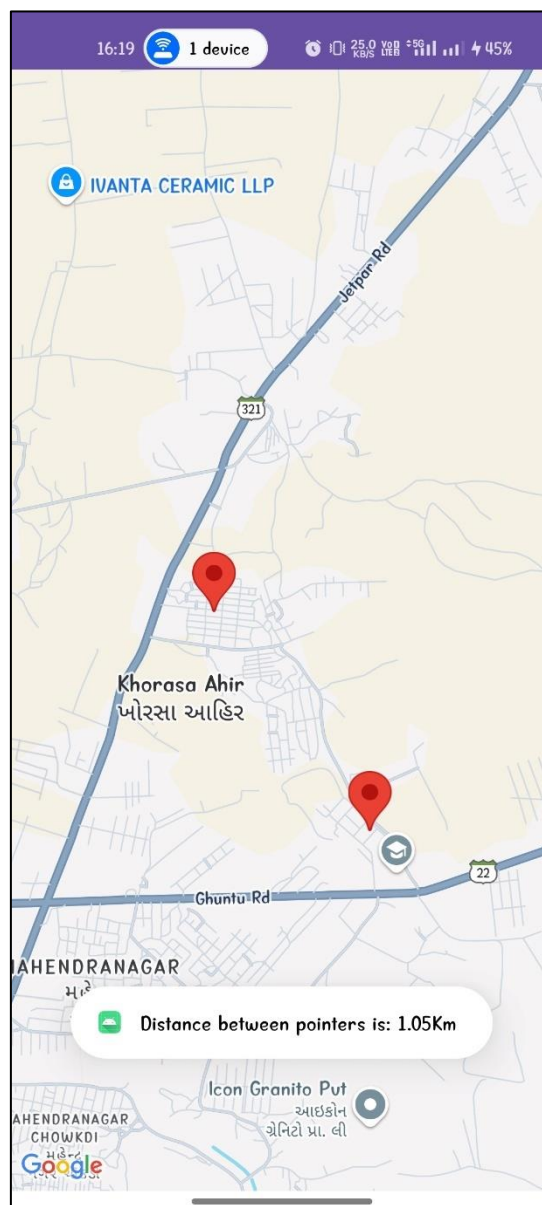
```

```

PackageManager.PERMISSION_GRANTED &&
ContextCompat.checkSelfPermission(this,
android.Manifest.permission.ACCESS_COARSE_LOCATION) !=
PackageManager.PERMISSION_GRANTED) {
    ActivityCompat.requestPermissions(this, new
String[]{android.Manifest.permission.ACCESS_FINE_LOCATION,
android.Manifest.permission.ACCESS_COARSE_LOCATION},
LOCATION_PERMISSION_REQUEST_CODE);
} else {
    Location location =
locationManager.getLastKnownLocation(LocationManager.GPS_PR
OVIDER);
    lastLat = location.getLatitude();
    lastLng = location.getLongitude();
    // Obtain the SupportMapFragment and get notified when the
map is ready to be used.
    SupportMapFragment mapFragment =
(SupportMapFragment) getSupportFragmentManager()
        .findFragmentById(R.id.map);
    mapFragment.getMapAsync(this);
}
}
}
}

```

Output:



Practical-24

AIM: Develop an application which performs Login using the Google account of the user.

Steps to follow:

1. Follow the steps from this video to connect with firebase for google login
2. <https://www.youtube.com/watch?v=YQ0fJUiOYbY>

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical24layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical24">

<TextView
    android:id="@+id/Practical24username"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="20dp"
    android:text="Username"
    android:textSize="24sp"

    app:layout_constraintBottom_toTopOf="@+id/Practical24txt"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent" />

<TextView
    android:id="@+id/Practical24txt"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login Using"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.5" />

<ImageButton
    android:id="@+id/Practical24login"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="32dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/Practical24txt"

    app:srcCompat="@drawable/common_google_signin_btn_icon_dark"
    />

</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;
import android.content.Intent;
import android.os.Bundle;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import com.google.android.gms.auth.api.signin.GoogleSignIn;
import com.google.android.gms.auth.api.signin.GoogleSignInAccount;
import com.google.android.gms.auth.api.signin.GoogleSignInClient;
import com.google.android.gms.auth.api.signin.GoogleSignInOptions;
import com.google.android.gms.common.api.ApiException;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import com.google.firebase.auth.GoogleAuthProvider;

public class Practical24 extends AppCompatActivity {

    private FirebaseAuth firebaseAuth;
    private GoogleSignInClient signInClient;
    private GoogleSignInOptions gso;
    private ImageButton btn;
    private TextView username,txt;
    FirebaseUser currentUser;

    // Register the launcher for Activity Result API
    private final ActivityResultLauncher<Intent> signInLauncher =
registerForActivityResult(
    new ActivityResultContracts.StartActivityForResult(),
    result -> {
        if (result.getResultCode() == RESULT_OK) {
            // Handle the Google Sign-In result
            Intent data = result.getData();
            handleSignInResult(data);
        } else {
            Toast.makeText(Practical24.this, "Sign-in failed",
            Toast.LENGTH_SHORT).show();
        }
    }
);

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```

EdgeToEdge.enable(this);
setContentView(R.layout.activity_practical24);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical24layout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
    return insets;
});

// Initialize Firebase Auth and Google Sign-In client
firebaseAuth = FirebaseAuth.getInstance();
gso = new
GoogleSignInOptions.Builder(GoogleSignInOptions.DEFAULT_SIGN_IN
)
    .requestIdToken(getString(R.string.default_web_client_id))
    .requestEmail()
    .build();
signInClient = GoogleSignIn.getClient(this, gso);

// Set up the login/logout button
btn = findViewById(R.id.Practical24login);
username = findViewById(R.id.Practical24username);
txt = findViewById(R.id.Practical24txt);

currentUser = firebaseAuth.getCurrentUser();

// Check if user is logged in
if (currentUser != null) {
    username.setText(currentUser.getDisplayName());
    txt.setText("Logout by clicking this button again");
    Toast.makeText(Practical24.this, "Logged in as: " +
currentUser.getEmail(), Toast.LENGTH_SHORT).show();
}

btn.setOnClickListener(v -> {
    currentUser = firebaseAuth.getCurrentUser();

    // Check if user is logged in
    if (currentUser != null) {
        // User is logged in, perform logout
        firebaseAuth.signOut();
        signInClient.signOut().addOnCompleteListener(task -> {
            Toast.makeText(Practical24.this, "Logged out successfully",
Toast.LENGTH_SHORT).show();
        });
    }
});

```

```

username.setText("Username");
txt.setText("Login Using");
} else {
    // User is not logged in, perform Google sign-in
    Intent signInIntent = signInClient.getSignInIntent();
    signInLauncher.launch(signInIntent); // Use the
ActivityResultLauncher to start sign-in
}
});
}

// Handle sign-in result
private void handleSignInResult(@Nullable Intent data) {
    try {
        GoogleSignInAccount account =
GoogleSignIn.getSignedInAccountFromIntent(data).getResult(ApiExce
ption.class);
        if (account != null) {
            // Firebase authentication with Google
            firebaseAuthWithGoogle(account.getIdToken());
        }
    } catch (ApiException e) {
        Toast.makeText(this, "Sign in failed",
Toast.LENGTH_SHORT).show();
    }
}

private void firebaseAuthWithGoogle(String idToken) {
firebaseAuth.signInWithCredential(GoogleAuthProvider.getCredential
(idToken, null))
    .addOnCompleteListener(this, task -> {
        if (task.isSuccessful()) {
            // Sign-in success, show a toast with the user's email
            (username)
                currentUser = firebaseAuth.getCurrentUser();
            if (currentUser != null) {
                Toast.makeText(Practical24.this, "Logged in as: " +
currentUser.getEmail(), Toast.LENGTH_SHORT).show();
                username.setText(currentUser.getDisplayName());
                txt.setText("Logout by clicking this button again");
            }
        } else {
            Toast.makeText(Practical24.this, "Authentication failed",
Toast.LENGTH_SHORT).show();
        }
    });
}
}

```

Output:

